L-PICK Controller

Command Reference

Ver.4.1

AIOI･SYSTEMS CO.,LTD.

History of Revisions

Date Printed	Version	File Name	Revision
	Ver.3.0		
2006 July 3	Ver.3.1	Comref_Ver.3.1	Revision of text, layout and charts
2006 Sep 20	Ver.3.2	Comref_Ver.3.2	Eliminated XW-P, changed the use and status of commands
2006 Dec 6	Ver.3.3	Comref_Ver.3.3	On the **FP1**, **FP5** command description [5] Display Data was changed from 5 digits to 6 digits. In the Note, 5 digits was changed to 6 digits.
2007 Mar 19	Ver.3.4	Comref_Ver.3.4	Revised "**D0**" of the "**Ar**" command. Updated the fixed value of the output data of the "**J**" command. Updated the fixed value of response command from the "**Ai**" command.
2007 July 10	Ver.3.5	Comref_V35	Updated the address of AIOI Systems USA Corp.
2008 May 23	Ver.3.6	ComrefV36	・changed the title from TW Series / XW Series to L-PICK Controller, 1.1 Overview ・NW Series added Modified: 1.1. Overview, 1.2 Words Used in the Manual (Ethernet Type), 1.3 How to Read Commands, 4. Commands, 4.4.1. Default, 5.3. PC's Resend Condition Added NW to the commands listed at top: @, Z, M, Am1, Am2, Am3, Am4, Am7, PP1, PP5, P1, P4, P5, L, D, t, O, N, PP2, P2, A, Az, Ac, ・1.3 How to Read Commands – Added the explanation of TW ・4.4.1. Default: XW Series functions added ・4.4.Operation mode specification commands Revised the title of the Am4 command. ・4.5. Operation Instruction Command Revised the explanation of the FP1, FP5 and Mode Array ・5.7.Number 0Light module. Deleted a phrase "Display only for the board type controller" under "Number of retries"
2009 Mar 24	Ver.3.7	ComrefV37	・Changed the NW marks ・Added NW297x series ・Added commands (**J**(16bit), **PP509**, **Ame**, **Amf**) ・Part of "**Z1"** command content changed ・Part of the "**Az**", "**t"** command added
2009 Mar 27	Ver.3.8	ComrefV38	Changed the applied timing of "**Ame**", ModeArray "**me**" commands
2016 Sep 16	Ver.4.0	ComrefV40	・Revised the related manuals ・Deleted the description on NW297x Series and NW294x Series ・Deleted the commands used only for the NW series. ・Deleted the description of **AiPlug** ・Added the table of contents for each commands category ・Revised the description of light module +/- key on the "**@"** command. Added notes. ・Revised the maximum data length of the "**D**" command ・Revised the title of the "**Am4**" command ・Revised the **Flash** specification of the "**Am1**" command (for batch display) ・Added **Flash at a high speed** of the "**Am1**" command (for batch display) ・Revised the **N** command description on the Ethernet

			・Revised the **Az** command description on response [3] Switch input status 1 ・Added a description on where to find troubleshooting information
2018 Apr 27	Ver.4.1	ComrefV41	・Added the quantity revision mode to the "**ma**", and "**Ama**" commands of ModeArray ・Deleted the **Ama** command used only for TW2060 ・Added the "**Ame**" command ・Revised the document

Related Manuals

- Application Development Guide
- Controller User's Manual
- TW Series Troubleshooting
- JW Series Troubleshooting

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1. Introduction

1.1. Overview

This manual is a command reference on common specifications for all standard L-PICK series controllers and page-turning controllers.

On specifications of the page-turning controller, refer to section **"3. Page-turning Controller**".

The controller sends commands through the application software to control modules. It is important to read section "**5. Supplement**" to understand how this is done.

L-PICK Series

TW Series
AW Series
DW Series
JW Series
SW Series
MWU Series

Wired
Unless a specific series is noted, **they are referred to as the "TW series"** from here on.

XW Series

NW297x Series → Refer to **NW Series Command Reference**

1.2. Words Used in the Manual

The meaning of words and expressions in this manual are shown on the chart.

Words	Definition
Ethernet type	Ethernet controller, Ethernet junction box (TW2161, TW2161P, TW2171, TW2208, XW37880)
Board type	Board type controller (TW2131, TW2131P, XW37870, XW37870P)
RS-232C type	RS-232C type controller (TW2111, TW2112, TW2113, TW2114, TW2115, XW37100, XW37120, XW37810)
Header	Command identification. First character of the command
Sub header	Second command identification. From the second character to the Address.
Parameter	Content of the command
JB	Abbreviation for junction box
BCRI/F	Abbreviation for barcode reader interface
RS-232CI/F	RS-232C interface
Module	General term for devices connected to the junction box.
LED	LED on the light module's **CONFIRM** button
SEG	Light module's 7-segment LED (displays quantity). Some do not have SEG.
BUZ	Light module sounds when an instruction is sent. There are light modules that do not give out sound.
+/- key	Keys to change the quantity. Some light modules do not have the +/- key. Also see → Quantity revision
Digit switch	Switches to specify possible addresses on each channel of board type controller. Does not exist on the Ethernet or RS-232C types.
TW2060 type	6-digit light modules, TW2060, TW2060-01
TW2414 type	Duct type alphanumeric batch display, 14 SEG.
Display status	LED, SEG, BUZ operation specification when light modules receive an instruction.
Operation instruction command	Commands for the controller to monitor the light modules.
"poka yoke" or error prevention	Prevention of human errors (poka). The poka yoke" light module prevents a worker From picking items from the wrong location.
Overwriting	The next operation instruction is sent before the current operation is terminated. Normally, until an operation is terminated, the next instruction command cannot be Sent to each light module. There are, however, commands that can be sent before terminating, such as **PP5, P5**, etc.
Quantity revision	The data displayed by the operation instruction can be revised using the **+/-** key or **Fn** key. After data has been revised and by pressing the **CONFIRM** button (operation completion), the application receives the revised quantity. If there is shortage of items during order accepting or picking, changes to the displayed quantity can be made by using the **+/-** key or **Fn** key.
Shortage	Lack of items to be picked or delivered due to insufficient items in stock during receiving and picking. By pressing the **Fn** key, then the **CONFIRM** button, "shortage" is reported to the application.

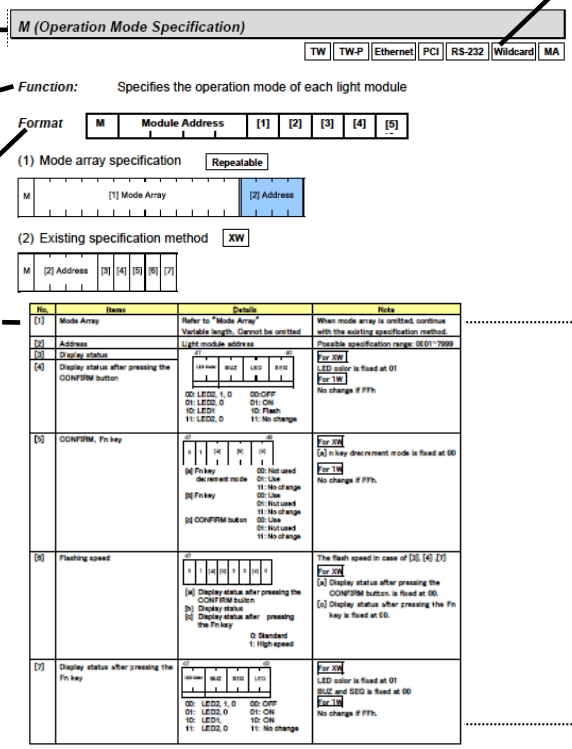
1.3. How to Read Commands

Title of the Command

Function
Explains command functions

Format
Command Format

Data in this section



Command Usage Conditions

Controller Series | Controller Type | Command Condition

TW:
TW series
AW series
DW series
array enabled
JW series
SW series

Ethernet
PCI
RS-232C

Repeatable: Can repeat
Wildcard: Can specify wildcard to the address
MA: Mode

TW-P: TW series page turning
NW4: NW294x Series
NW7: NW297x Series
XW: XW Series

Details of Command Format

Details of specified value
Always read Notes

Maximum Command Length

Presented when part of the command is repeatable or variable

Command Example

Example of command specifications
Read the explanation.

Notes

Always read the command notes.

Other

Operation

Explains the operation when there is a change in the light module operation after the command is sent.

Response

Response after the command is sent. An explanation is given when a response after a command is sent is different from that of the Section 2.2.

2. Communication

2.1. Command Format

The following is the format when a command is sent to each type of controller.

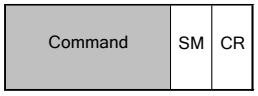
Ethernet controller



Board type controller

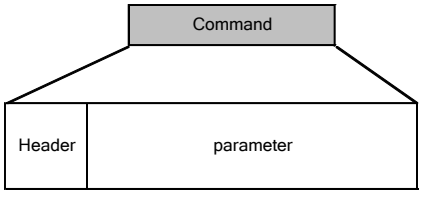


RS-232C type controller



Command	**Command**
STX	Beginning of the command: 02h
ETX	At the end of the command: 03h
Seq No.	Sequence number: 000 ~ 999
Data Length	Command length: 1 ~ 1015 (byte)
SM	Checksum
CR	End of command: 0Dh

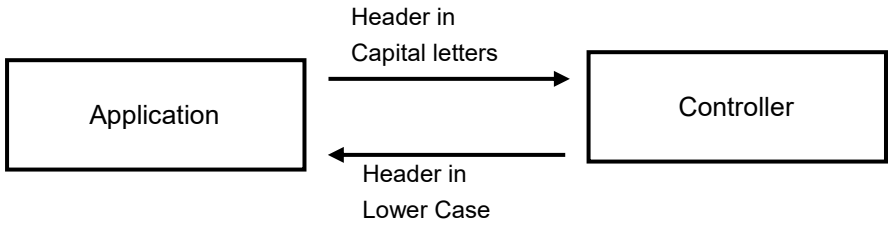
The header of the command that is sent to the controller must always be in upper case letters. Some commands are composed of HEADER + Sub Header + Parameter. For details, refer to Section 4. Details of the Commands.



2.2. Sending a Command and Getting a Response

When the controller receives a command from the application, it always sends a response. When the application sends a command, it always receives a response before sending the next command.

In the case of the barcode data and light module completion for the Ethernet type controllers the controller will send a command. The application must always send a response.



Controller's Response:

Command Accepted

o

Command rejected

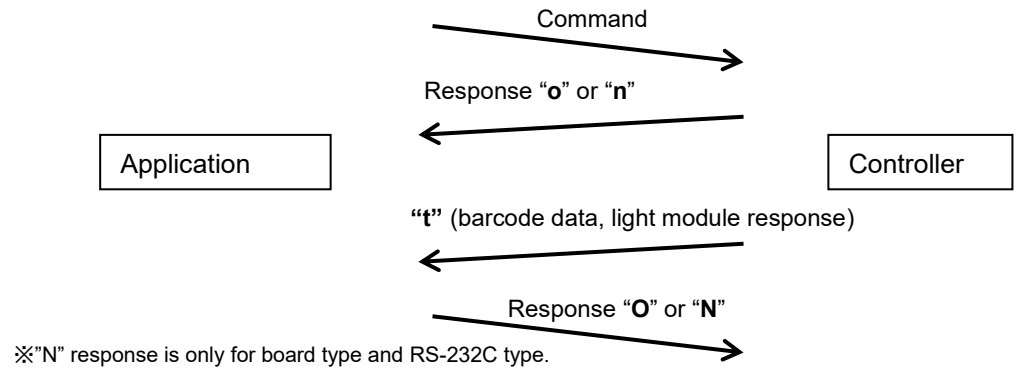
n

Reasons for command rejection:

(1) There is a mistake in the header.
(2) If there is a sub header, there is a mistake in the sub header.
(3) The address that is specified is out of range of that module.
(4) The address specified by the digit switch (SW2-SW3) located on the side of the controller is out of range (for the board type only).
(5) The specified number of characters is more than the assigned parameter. If that happens, controller will process only up to the assigned number of the parameter.
(6) There is an error in the checksum (for RS-232C type controller only)

Example of Sequence:

Basic sequence between the application and controller is illustrated below. For other examples, refer to the "Controller User's Manual."



※"N" response is only for board type and RS-232C type.

2.3. Waiting Time for a Response

Ethernet Type	**More than 5 seconds**
Board Type / RS-232C Type	**More than 3 seconds**

Normally a response is always returned from the controller. If there is an abnormal light module, a response to confirm abnormality can take 3 seconds. Usually a response is immediate, but there may be a delay.

3. Page-Turning Controller

This controller can perform an operation called “Page-turning” and a letter “P” is at the end of model Number (TW21××P)

3.1. What is Page-Turning?

The controller sends a command to a light module with maximum of 10 operation instructions in advance, and the application obtains the result after completing all instructions. The operation instructions that were sent in advance are called “pages.” The controller performs the “page-turning” (switching displayed data) after the **CONFIRM** button is pressed. This is the function of the special “page-turning” controller.

3.2. Specifications and Limitations of the Page-Turning Controller

- Use the special commands (**Pm**, **P6**) for page-turning function. The rest of the commands are the same as those of the standard controller (*1).
- Can return to the previous page (set up with the **“@”** command).
- Can use the +/- key operation (set up with the **“@”** command).

*1：Limitation on the number of light modules that works during operation.

The number of light modules that works during operation.	**1000 units**
Within 1,000 units, the number of light modules that works with the page-turning function using the Pm, P6 commands during operation.	**600 units**

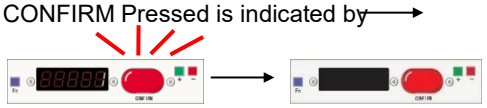
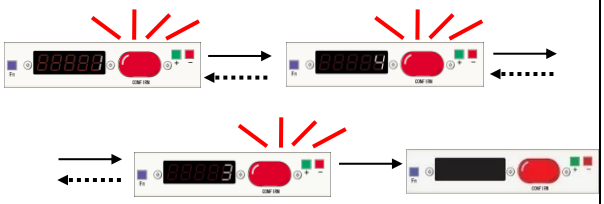
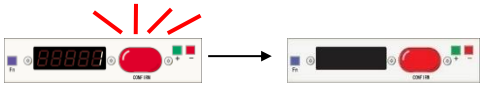
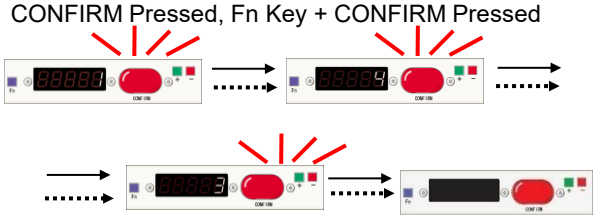
3.3. +/- Key

In order to change quantity or display the previous page, set the “**@**” command to “Use +/- key”. The functions are shown below for the “**@**” command, “Use +/- key”.

Command	Function
Operation instruction commands (**PP1, PP5** etc.)	Revise quantity
Special page-turning operation instruction (**Pm**, **P6**)	Display previous page

For details, refer to “**@**” command in Section 4.3. Initialization Setup Command.
The operation and response from the controller with the “**@**” command are shown on the chart on the next page.

The Operation and Response from the Controller with the "@" Command Setup

<table>
<tr><th></th><th>Command</th><th>Operation</th><th>Response Code (∗1)</th></tr>
<tr><td rowspan="3">When the +/– key is used for the Setup (@0001)</td><td rowspan="2">Operation Instruction Commands (PP1, PP5, etc.)</td><td>CONFIRM Pressed is indicated by
  </td><td>00
Used the +/– key:
00<xxxxx> (∗2)</td></tr>
<tr><td>Fn Key + CONFIRM Pressed is indicated by
  </td><td>10</td></tr>
<tr><td>Page-turning Commands (Pm, P6)</td><td>CONFIRM Pressed, Fn Key + CONFIRM Pressed
  </td><td>00
Used the +/– key:
00<x-xxxxx> (∗2)</td></tr>
<tr><td rowspan="3">When the +/– Key is not used for the setup (@00000)</td><td rowspan="2">Operation Instruction Commands (PP1, PP5, etc.)</td><td>CONFIRM Pressed
  </td><td>00</td></tr>
<tr><td>Fn Key + CONFIRM Pressed
  </td><td>10</td></tr>
<tr><td>Page-turning Commands (Pm, P6)</td><td>CONFIRM Pressed, Fn Key + CONFIRM Pressed
  </td><td>CONFIRM Pressed
00

Fn Key +CONFIRM Pressed
00<x-00000 > (∗3)</td></tr>
</table>

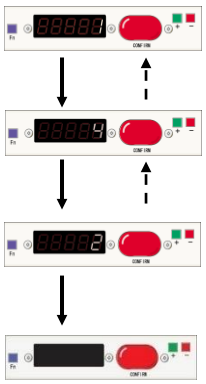
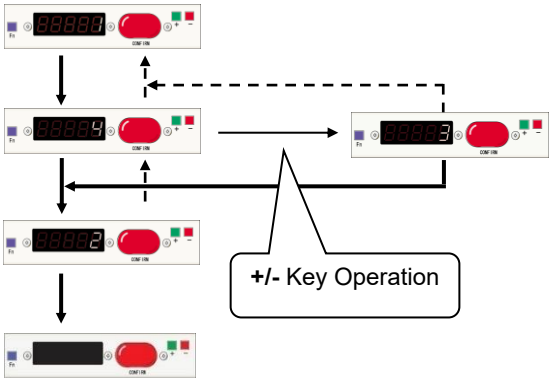
∗1: The "**t**" command code is for the Ethernet type. The "**h**" code, responding to the "**H**" command, is for the board type and RS-232C type.

∗2: The "xxxxx" will change depending on the display data.

∗3: The normal completion code and display data 00000 will be returned.

The examples of the operation and response code when the page-turning commands (**Pm**, **P6**) are used for 3 pages of operation instructions.

Using the +/- Key

CONFIRM press is indicated by ⟶ **Fn** Key + **CONFIRM** is indicated by ⋯⋯▸ (Without **+/-** Key Operation)	**CONFIRM**, **Fn** Key + **CONFIRM**, **+/-** Key Operation
	
Response Code 00	Response Code 00<1-△△△△3> (✱4)

✱4 : “△” indicates space (20h).

4. Commands

4.1. Commands At-A-Glance

Header	Details	LM	BD	SL	OI/O	BCRI/F	RS-232CI/F	DI/O
@	Specify controller operation							
Z	Initialization setup	○	○	○	○	○	○	○
Z1	Initialization	○	○	○	○	○	○	○
B1	Auto completion mode setup	○						
M	Operation mode setup	○	○	○	○			
Am *1	Operation mode setup	○	○	○	○			
FAm	Operation mode setup	○*2						
PP1	Can specify display data after pressing the Fn key Give block operation instruction	○						
PP5	Rewritable. Can specify display data after pressing the Fn key. Give block operation instruction	○						
P1	Give block operation instruction	○						
P4	Give block operation instruction with the mode specified.	○						
P5	Give block operation instruction with the mode specification. Rewritable.	○						
L	Specify individual operation	○						
FP1	Give block operation instruction	○*2						
FP5	Give block operation instruction with the mode specification. Rewritable	○*2						
Pm	Page-turning operation instruction with mode array	○						
P6	Page-turning operation instruction	○						
H *4	Check the completion status							
D	Terminate operation instruction	○	○	○	○			
X	Turn ON the batch display to show message		○					
Q2	Turn ON the batch display to show message		○					
T	One-point I/O ON/OFF				○			
T	Signal light ON/OFF			○				
G	For BCRI/F and RS232CI/F					○	○	
t	Report from the controller	○			○	○	○	
Ar	Check RS232CI/F						○	
J	Digital I/O Output							○
E	Digital I/O Input							○

Ai	Read digital I/O output data							○
PP2	Display module	○	○	○	○			
P2	Display module	○	○	○	○			
FP2 *2	Display module	○	○	○	○			
A	Maintenance specification	○	○	○	○	○	○	
Az	Inquiry of module status	○		○	○	○	○	○
Ac	Module address setup	○	○*3	○	○	○	○	
O	Response from the application (ACK)							
N *4	Denial from the application (NACK)							

○: Applicable TW series・XW series

LM: Light module
BD: Batch display
SL: Signal light
OI/O: One-point I/O
BCR I/F: Barcode reader interface
RS-232C I/F: RS-232C interface
DI/O: Digital I/O

∗1: The "**Am8**" command is a special command for the TW2916 error prevention light module "poka yoke." The "**Ama**" and "**Amb**" commands are special commands for the TW2060. The "**Amd**" command is only applicable for the TW2414.
∗2: Special commands for the TW2060 type.
∗3: Can only be used for the duct type batch display (TW2414, TW2417, etc.)
∗4: Special command for the board type and RS-232C type.

4.2. What to Watch for When Writing Commands

Do not include the values shown below in the parameter.

Ethernet type / PCI board type	**02h, 03h**
RS-232C type	**0Dh**

4. .1. Address Specification with Wildcard

A wildcard, "**?"**, can be used to specify the addresses.

Command Example: To turn off the light module addresses from 1~999.

D	Address			
D	0	?	?	?

Command Example: To turn off the light module addresses 1000, 1010, 1020, 1030, 1040, 1050, 1060, 1070, 1080, 1090

D	Address			
D	1	0	?	0

Note

The specification range depends on the command. When specifying a wild card, only specify within the allowable range for this command.

Address Specification Commands At-A-Glance

Header, Sub-Header	Description	Wildcard Specification (Yes / No)
M	Operation mode specification	Yes
Am	Operation mode specification	Yes
FAm	Operation mode specification	Yes
PP	Block operation instruction with mode array	No
P1	Block operation instruction	No
P4	Block operation instruction with operation mode	No
P5	Block operation instruction with operation mode. Overwritable.	No
L	Standard operation instruction	No
Pm	Page turning operation instruction with mode array	No
P6	Page turning operation instruction	No
FP1	Block operation instruction	No
FP5	Overwritable block operation instruction with operation mode	No
H	Completion confirmation	No
D	Operation instruction termination	Yes
Am	Batch display operation mode specification	Yes
X	Turn on the batch display	Yes
Q2	Turn on the batch display	Yes
M	One-point I/O operation mode specification	Yes
Am	One-point I/O operation mode specification	Yes
T	One-point I/O ON/OFF	Yes
M	Signal light operation mode specification	Yes
Am	Signal light operation mode specification	Yes
T	Signal light ON/OFF	Yes
Az	Inquiry of module status	No
Ac	Module address specification	No
G	For BCRI/F and RS-232CI/F	Yes
Ar	Check RS-232CI/F	No
J	Digital I/O output	Yes
E	Digital I/O input	No
Ai	Read digital I/O output data	No
P2	Display module	Yes
FP2	Display module	Yes

4.3. Initialization Setup Command

Initialization command at-a-glance

Command	Description	Applicable		Page No.
		TW	XW	
@	Operation specification to the controller	○	○	15
Z	Initialization setup	○	○	17
Z1	Initialization	○		18
B1	Automatic completion mode setup	○		19

○: Applicable

@ (Operation Specification to the Controller)

TW | TW-P | XW | Ethernet | PCI | RS-232C

Function

Operation instruction to the controller

Format

TW-P

@	[1]	0	[2]	0	[3]

TW | XW

@	[1]	0	[2]	0	0

Response

@	[4] Version	9	9	[5]	[6]

No.	Items	Details
[1]	The number of attempts the controller tries to send a command to a module.	0: 5 times (default) 1: 0 time PCI
[2]	Modules that the controller turns OFF when the entire operation within a block terminates.	0: Signal light (default) 1: Batch display Ethernet PCI
[3]	+/- key of the light module	0: Not use (default) 1: Use
[4]	Version	Version of the controller
[5]	Upper limit of the light module address in Channel 2	PCI Digit switch value RS-232C, Ethernet "9"
[6]	Upper limit of the light module address in Channel 1	PCI Digit switch value RS-232C, Ethernet "7"

Command Example

@	[1]	0	[2]	0	[3]
@	0	0	1	0	0

@	[4] Version		9	9	[5]	[6]
@	1	0	9	9	9	7

Explanation

[1] Attempts or retries: **5 times**

[2] When an operation within a block is terminated: **Batch display is turned OFF**

[3] The +/- key of the light module: **Not use**

[4] Version**1.0**

[5] The digit switch setup for channel 2: **9**

[6] The digit switch setup for channel 1: **7**

Note

(1) Whenever the initialization setup command is needed, send the "**@**" command. After sending, always send the "**Z**" command.

(2) Regarding **TW** and **XW**, there is no function to set the **+/-** key of the light module not to be used.

Z (Initialization Setup)

TW | TW-P | XW | Ethernet | PCI | RS-232C

Function

The "**Z**" command initializes devices and the controller.

Format

Z

Operation

(1) Initializes the operation mode of each module or device.

(2) Refer to Section "**4.4.1. Default**" for the initialization value of each module.

(3) When using the Ethernet controller, reset the Seq. No. of the "**t**" command the controller sends to "000".

Note

At the start of the operation, always send this "**Z**" command.

Z1 (Initialization)

TW | TW-P | Ethernet | PCI | RS-232C

Function

Initializes all modules and controllers

Format

Z	1

Operation

Devices	Operation
Light module	Reset the hardware. The light modules that are lit up will be turned OFF. There will be no report on abnormal light modules after the reset.
Barcode reader interface	Reset the hardware. Those that are open will be closed. After the reset, there will be no malfunction report.
Batch display (duct type only)	Reset the hardware. The output will be all OFF.

Note

(1) When device malfunctioning is detected, it may be possible to restore (only light modules or barcode reader interface). However, it cannot be restored if there is a breakdown of a device or contact failure.

(2) The process time is about 1.5 seconds. Allow 1.5 seconds or more for the next command to be sent.

B1 (Automatic Completion Mode Setup)

TW | Ethernet PCI RS-232C

Function

Start / Stop the automatic completion mode (used for the operation test)

Format

B	1	[1]	[2]	[3]

No.	Name	Details	Note
[1]	Start/stop automatic completion	0: Stop 1: Start	When power is supplied to the controller, it is 0: Stop
[2]	Specified time to end the automatic completion.	・The amount of time (secs) it takes to complete according to the completion type [3].) ・The specification range is 1~127 seconds	・For 1~99, use “0” to fill unused digits in front. (e.g., 5 secs= 005 secs) ・If more than 127secs, force it to 127secs.
[3]	Types of completion	0: Normal completion 1: Shortage 2: Light module abnormal	When “light module abnormal” is set up, no output to the “0” light module or error log.

Command Example

B	1	[1]	[2]			[3]
B	1	1	0	0	5	0

Explanation

[1] Automatic completion, Start / Stop: **Start**
[2] Setup time for automatic completion: **5 seconds**
[3] Completion type: **Normal completion**

Note

(1) This command is used for an operation test. Do not use it during operation.
(2) For details of automatic completion mode function, refer to the following page.

Automatic Completion Mode

■Summary of Automatic Completion Mode Functions

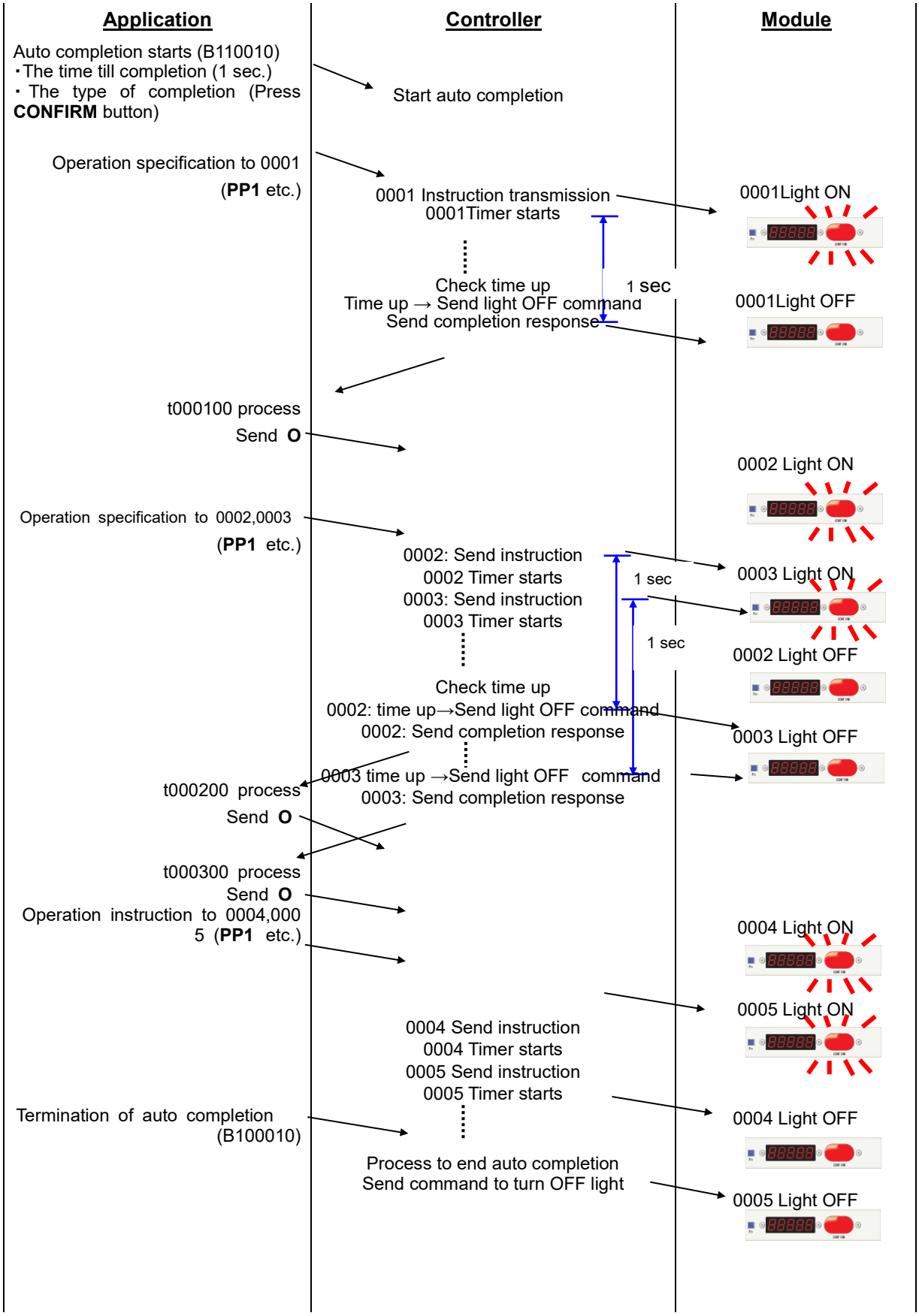
This function enables the controller to perform completion without pressing the **CONFIRM** button of specified light modules with a given instruction. In the application, specify the Start/Stop of an automatic operation, the required time until completion, completion types (normal, shortage, or malfunction). The automatic completion is used to conduct operation tests. Do not use it during actual operation.

(1) Applies to the light modules that are specified by an operation command to perform the automatic completion from “Start” to “Stop”.
(2) The automatic completion time is measured from the time when the controller receives an operation instruction until that light module completes the operation.
(3) All light modules complete the automatic completion according to 3 categories (types) (normal completion, shortage, abnormal).
(4) The completion report is the same as the actual operation.

Operation	Ethernet	PCI RS-232C
Automatic Completion	“**t**” command	“**h**” command
Abnormality occurs within time allowed for auto completion.	“**t**” command	“**h**” command
Press the **CONFIRM** button within time allowed for automatic completion	“**t**” command	“**h**” command
Automatic completion command	Light module OFF, No reporting	Light module OFF, No reporting

(5) If an automatic completion is abnormal, there is no output in the “0” light module or in an error log.
(6) Previously specified “use/not use” of the **CONFIRM** button or **Fn** key (“**M**” or “**Am4**” commands are invalid)

■Operation Flow



■Operation Details

The operation instruction command with mode specified will work according to the mode specification of the **CONFIRM** button, **Fn** key, and **Fn** decrement mode. The details of the mode and automatic completion operation are shown below.

Mode Specification			**Automatic Completion Type**		
Fn Decrement Mode	**Fn**	**CONFIRM**	Completion	Shortage	Abnormal
Valid	Valid	Valid	○	×	○
		Invalid	×	×	○
		No Change	―	×	○
	Invalid	Valid	○	×	○
		Invalid	×	×	○
		No Change	―	×	○
	No change	Valid	○	×	○
		Invalid	×	×	○
		No Change	―	×	○
Not Valid	Valid	Valid	○	○	○
		Invalid	×	×	○
		No Change	―	―	○
	Invalid	Valid	○	×	○
		Invalid	×	×	○
		No Change	―	×	○
	No change	Valid	○	―	○
		Invalid	×	×	○
		No Change	―	―	○
No Change	Valid	Valid	○	―	○
		Invalid	×	×	○
		No Change	―	―	○
	Invalid	Valid	○	×	○
		Invalid	×	×	○
		No Change	―	×	○
	No Change	Valid	○	―	○
		Invalid	×	×	○
		No Change	―	―	○

○: Automatic Completion　x: No Automatic Completion　―: Cannot be determined – can be either ○ or x

■Restrictions

- When automatic completion is performed, the display condition after the **CONFIRM** button is pressed will not work correctly. Everything will turn OFF, If the **CONFIRM** button is manually pressed before the automatic completion is completed, the display condition is “Valid”.
- The automatic completion mode will work only with the standard TW controller versions listed below or later versions.

Controller Type	**Version**	**ROM No.**
Ethernet	V3.8	1121
PCI	V5.0	1134
RS-232C	V3.3	1133

4.4. Operation Mode Specification Command

Various operations, such as turning ON or OFF the module's light or buzzer can be specified Using the commands listed below.

A Summary of Operation Mode Specification Commands

Command	Setup Details	For		Page No.
		TW	XW	
M	[For mode array] Can make a combination of **Am1~Am4, Am8, Ama, Amb**	○		26
	[For fixed length format] ・ Display status ・ Display status after pressing the **CONFIRM** button ・ Display status after pressing the **Fn** key ・ **CONFIRM** button, **Fn** key specification ※The full-color light modules cannot display all colors.	○	○	31
Am1	Display status	○		33
Am2	Display status after pressing the **CONFIRM** button	○		35
Am3	Display status after pressing the **Fn** key	○		37
Am4	Specification of the **CONFIRM b**utton**, Fn** key	○		39
Am7	Specification of display data after pressing the **Fn** key	○		40
FAm7	Specification of display data after pressing the **Fn** key (for TW2060 only)	○		41
Am8	Setup of "poka yoke" mode (error prevention mode)	○		42
Ama	Specification for using the **DigNum** key	○		44
Amb	Scroll setup condition by pressing the **Fn** key (for TW2060 only)	○		45
Amd	Extend operation mode	○		46
Ame	Display condition setup for each SEC of electronic paper display	○		48

○ : Applicable

When using the TW series, it is recommended that you sue the "**M**" command (mode array) for all specifications except "display data after pressing the **Fn** key." For more information on the mode array, refer to Section "4.4.2 Mode Array".

Note

(1) To find out the relationship between the LED number and colors, refer to Section "5.5 Light Module LED Number and Color Relationship."

(2) "No change" means there is no change from the previously setup. Refer to the following page, Section "4.4.1. Default" for default operation of various devices.

(3) The display status after pressing the **Fn** key is the status when only the **Fn** key is pressed. If the **CONFIRM** button is pressed after pressing the **Fn** key; **BUZ, LED, SEG** will all turn OFF.

4.4.1 Default

(1) Default Operation Mode of the TW Series Light Modules

	LED0	**LED1**	**LED2**	**SEG**	**BUZ**
Display status	Flash	Flash	Flash	ON	Flash
Display status after pressing the **CONFIRM** button	OFF	OFF	OFF	OFF	OFF
Display status after pressing the **Fn** key	ON	ON	ON	ON	OFF
CONFIRM button, **Fn** key	Use				
Flashing speed	Standard				
Fn key decrement mode *1	Not used				
Display data after pressing the Fn key	-----				

Default Operation Mode of the XW Series Light Modules

	LED	**SEG**	**BUZ**
Display status	Flash	ON	Flash
Display status after pressing the **CONFIRM** button	OFF	OFF	OFF
Display status after pressing the Fn key	OFF	Flash	OFF
CONFIRM button, **Fn** key	Use		
Flashing speed	Standard		

(2) Default Operation Mode of the TW Series “Poka Yoke” light module

“Poka yoke” action	Use
Automatic **CONFIRM**	Use
Automatic **CONFIRM** timing	When a hand is withdrawn from a location
Sensor’s reaction time when a hand is withdrawn from a location	0 sec
Sensor’s reaction time when a hand enters a location	0 sec

(3) Operation Mode of the TW2060 type of Light module at Default

Specify quantity revision key	DigNum key
Scroll mode	No scroll
Scroll speed	Standard

(4) Default Operation Mode of theTW2050FK Light Module

Specify quantity revision key	**DigNum** key

(5) Operation Mode of the TW Series Batch Displays at Default

Display (SEG)	ON
Scroll function	None
Left justified display *3	Use
Setup number of 14 SEG *3	12 characters

Default Operation Mode of the XW Series Batch Display

Display (SEG)	ON
Scroll function	None

(6) Default Operation Mode of the TW Series Signal Light (Tower)

Light (LED0)	ON

Default Operation Mode of the XW Series Signal Light (Tower)

Light (LED0)	ON

(7) Default Operation Mode of the TW Series Intelligent Signal Light (Tower)

Light (LED0)	ON
Buzzer (BUZ)	OFF

Default Operation Mode of the XW Series Intelligent Signal Light (Tower)

Light (LED0)	ON
Buzzer (BUZ)	OFF

(8) Default Operation Mode of One-Point I/O

OUT(LED2)	Flash

When “ON” is outputted to the one-point I/O, specify the display of LED0, LED1, and LED2 to “ON” with the “Am1” or “M” command.

*1 **Fn** key decrement mode

Note: The XW Series does not have this function.

Two ways of using the **Fn** key

(1) Default

- When the **Fn** key is pressed, the data will display “— — — — —” and when the **CONFIRM** button is pressed, it indicates “shortage” (10).
- When the Fn key is pressed again, the display data goes back to the original.

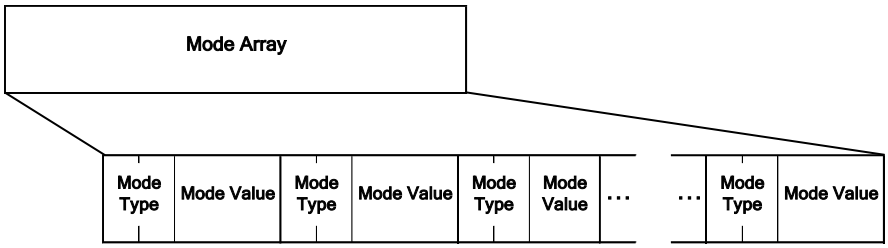
(2) **Fn** key decrement mode (operation mode of “**M**” / “**Am4**”)

- Decrease the display data by 1 (quantity revision).
  When the **CONFIRM** button is pressed, the display data will show normal termination “**aaaa00<nnnnn>**”.
- Press the **Fn** key when the displayed data is “**0**”, then the display data will return to the original value.

*3 Applicable for the TW2414 (Ver.1.5 or later). Only for batch displays.

4.4.2 Mode Array

With the Mode Array, the existing operation modes can be combined and be used inside a command. Because several operation modes can be sent as 1 command, the number of commands is reduced.

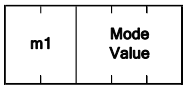


Note: The XW series does not have this function.

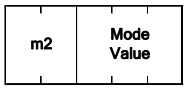
Mode array can specify the following commands:
Operation mode command: **M**
Operation instruction commands: **PP1, PP5, FP1, FP5, Pm**
Display commands : **PP2, FP2**

The mode type is an operation mode identifier. The mode value is a setup value.
The mode types are as follows:

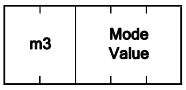
- Display status



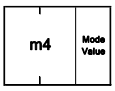
- Display status after pressing the **CONFIRM** button



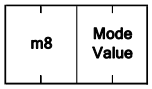
- Display status after pressing the **Fn** key



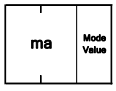
- **CONFIRM** button, **Fn** key



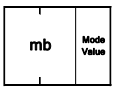
- Setup function of “poka yoke” light module



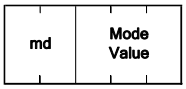
- Use / Not use of the quantity revision key



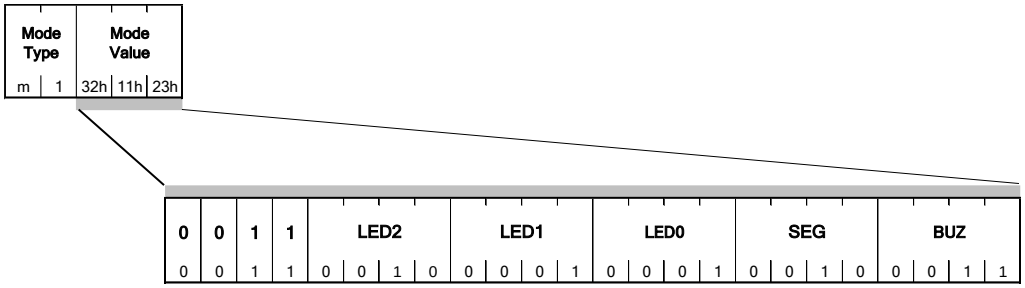
- Scroll setup condition after pressing the **Fn** key
  * for the TW2060 only



- Specification of expand operation mode



(1) "**m1**" (display status), "**m2**" (display status after pressing the **CONFIRM** button), "**m3**" (display status after pressing the **Fn** key)



Mode Value

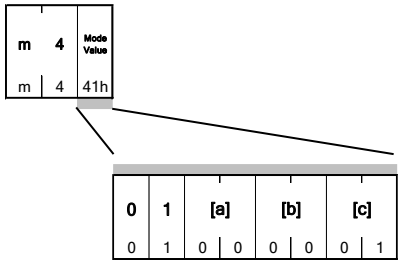
LED2	0001 : OFF
LED1	0010 : ON
LED0	0011 : Flash
SEG	0100 : Flash at a high speed
BUZ	0111 : No change

Explanation

Display status specification

LED2: **ON**, LED1: **OFF**, LED0: **OFF**, SEG: **ON**, BUZ: **Flash**

(2) "**m4**" (**CONFIRM** button, **Fn** key)



Mode Value

[a]	**Fn** key decrement mode	00 : Not use 01 : Use 11 : No change
[b]	**Fn** key	00 : Use 01 : Not use 11 : No change
[c]	**CONFIRM** button	

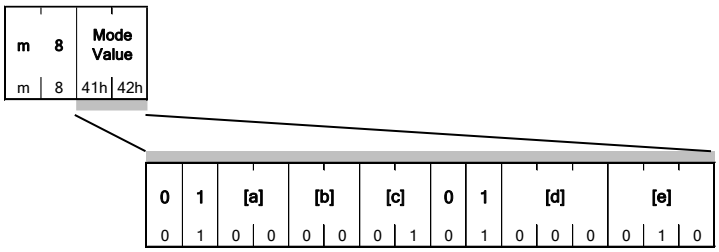
Explanation

[a] Fn key decrement mode: **Not used**

[b] Fn key: **Used**

[c] CONFIRM button: **Not used**

(3) "**m8**" (set up "poka yoke" or an error prevention function)



Mode Value

[a]	"poka yoke" action	00: Use 01: Not use 11: No change
[b]	Automatic **CONFIRM** button	00: Use 01: No use 11: No change
[c]	Automatic **CONFIRM** button timing	00: When a hand retracts from a location 01: When a hand enters a location1 11: No change
[d]	Response time of the sensor when a hand retracts from a location	000: 0 Sec 001: 1 Sec 010: 2 Sec 011: 3 Sec 111: No change
[e]	Response time of the sensor when a hand enters a location	000: 0 Sec 001: 0.1 Sec 010: 0.2 Sec 011: 0.5 Sec 111: No change

Explanation

[a] "poka yoke" action: **Use**

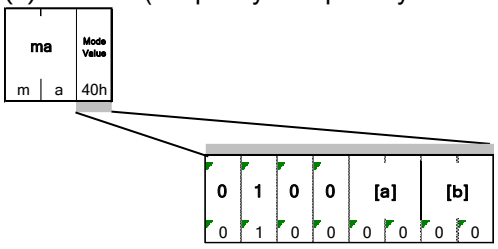
[b] Automatic CONFIRM button: **Use**

[c] Automatic CONFIRM button timing: **When a hand enters a location**

[d] Response time when a hand retracts from a location: **0 sec**

[e] Response time when a hand enters a location: **0.2 sec**

(4) "**ma**" (to specify the quantity revision key)



Mode Value

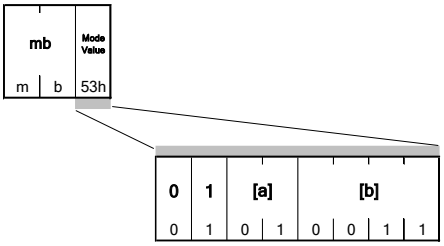
[a]	Specify the quantity revision mode	00: Normal (Space is not treated as a number value) 01: Special (Space is treated as a number value)
[b]	Specify the quantity revision key	00: DigNum key 01: Not use 10: +/- key 11: No change

Explanation

[a] Quantity revision mode: **Normal**

[b] Quantity revision key: **DigNum key**

(5) "**Mb**" (scroll setup condition after pressing the **Fn** key) (Specification of quantity revision key) [for TW2060 only]



Mode Value

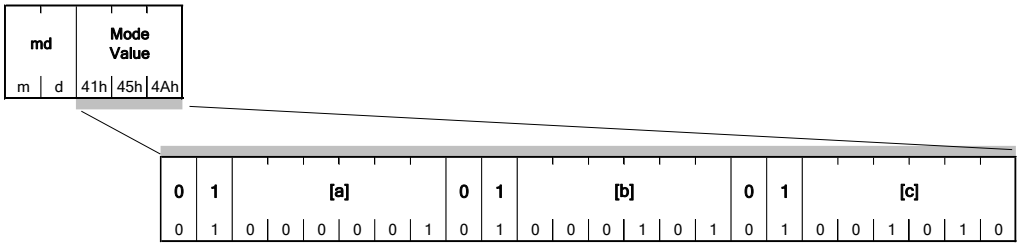
[a]	Scrolling mode	00 : No scroll 01 : Scroll ON 10 : Scroll ON (with a pause) 11 : No change
[b]	Scrolling speed	0001 : Standard 0010 : High speed 0011 : Fastest speed 1111 : No change

Explanation

[a] Scroll mode: **Scroll ON**

[b] Scroll speed: **Fastest**

(6) **"Md"** (specification of expand operation mode)



Mode value

[a]	Operation Number	000001
[b]	Expand operation mode	0 0 *A *B ∗A: Scrolling pauses with Q2, X command 00: Not used 01: Used ∗B: Left justified 00: Not used 01: Used
[c]	Expand operation mode	0 0 *A ∗A: Specify the number of 14 SEG 0100: 4 1001: 9 0101: 5 1010 10 0110 6 1011: 11 0111: 7 1100: 12 1000: 8

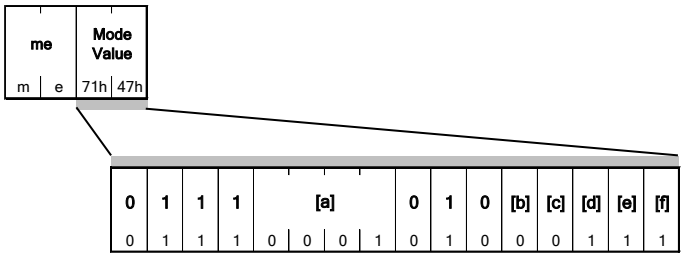
Explanation

[a] Operation number: **Expand the TW2414 operation**

[b] Scrolling pauses after the "Q2" or "X" command: **Used**

[c] Number of 14SEG: **10**

(7) "**me**" [SEG (digits) Normal / Reverse Display Condition]



Mode Value

[a]	Applied timing	0001 : When an operation instruction is given / Reverse display 0010 : After pressing the **CONFIRM** button / Reverse display 0011 : After pressing the Fn key / Reverse display 0100 : When an operation instruction is given / Small font display 0101 : After pressing the **CONFIRM** button / Small font display 0110 : After pressing the Fn key / Small font display	
[b]~[f]	Set up the SEG (digit) Normal/Reverse	From far right: [b]5th digit [c]4th digit [d]3rd digit [e]2nd digit [f]1st digit	0 : Normal 1 : Applied timing operation

Explanation

[a] Applied timing: **When an operation instruction is given**

[b] ~[f] SEG (digit) Normal /Reverse: **Only the last 3 digits are reverse display**

M (Operation Mode Specification)

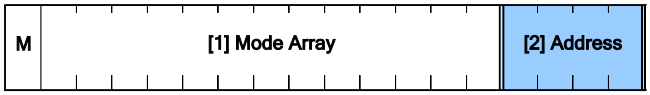
TW | TW-P | Ethernet | PCI | RS-232C | Wildcard | MA

Function

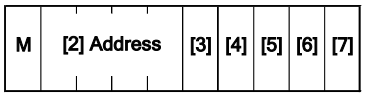
Specifies the operation mode of each light module

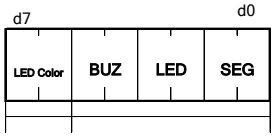
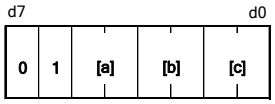
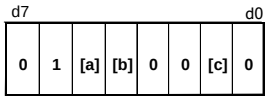
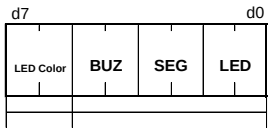
Format

(1) Mode array specification **Repeatable**



(2) Fixed length format specification method **XW**



No.	Items	Details	Note
[1]	Mode Array	Refer to "Mode Array" Variable length, cannot be omitted	When mode array is omitted, continue from [Address]
[2]	Address	Light module address	Possible specification range: 0001~7999
[3]	Display status	 00: LED2, 1, 0 00 : OFF 01: LED2, 0 01 : ON 10: LED1 10 : Flash 11: LED2, 0 11 : No change	For **XW** LED color is fixed at 01 For **TW** No change if FFh
[4]	Display status after pressing the **CONFIRM** button		
[5]	**CONFIRM** button, Fn key	 [a] **Fn** Key decrement mode 00: Not used 01: Used 11: No change [b] Fn key 00: Used 01: Not used 11: No change [c] CONFIRM key 00: Used 01: Not used 11: No change	For **XW** [a]Fn key decrement mode is fixed 00 For **TW** No change if FFh
[6]	Flashing speed	 [a] Display status after pressing CONFIRM button [b] Display status [c] Display status after pressing he Fn key: 0: Standard,1: High speed	The flash speed in case of [3], [4], [7] For **XW** [a] Display status after pressing the **CONFIRM** button is fixed at 0. [c] Display status after pressing the **Fn** key is fixed at 0.
[7]	Display status after pressing the Fn key	 00: LED2, 1, 0 00 : OFF 01: LED2, 0 01 : ON 10: LED1 10 : Flash 11: LED2, 0 11 : No change	For **XW** LED color is fixed at 01 BUZ and SEG is fixed at 00 For **TW** No change if FFh.

Maximum Command Length

(Mode Array)

Ethernet type	1015 bytes
Board type	1024 bytes
RS-232C type	

Command example

(1) Mode array, Repeatable

M	**[1] Mode Array**												**[2] Address**				**[2] Address**			
M	m	1	33h	33h	21h	m	2	32h	22h	21h	m	4	44h	0	0	0	1	0	0	0	2

Explanation

[1] Display status : LED2: **Flas**h, LED1: **Flash**, LED0: **Flash**, SEG: **ON**, BUZ: **OFF**
Display status after pressing the CONFIRM button : LED2: **ON**, LED1: **ON**, LED0: **ON**, SEG: **ON**, BUZ:**OFF**
CONFIRM button, Fn key : Fn key decrement mode: **Not used**, Fn key**: Not used**, CONFIRM button: **Use**

[2] Light module address: **0001, 0002**

(2) Fixed length format, operation mode specification with wildcard

M	**[2] Address**				**[3]**	**[4]**	**[5]**	**[6]**	**[7]**
M	1	?	?	?	05h	01h	40h	40h	FFh

Explanation

[2] Light module addresses: **1000～1999**

[3] Display status: LED color LED2, LED1, LED0, BUZ: **OFF.** LED: **ON**, SEG: **ON**

[4] Display status after pressing the CONFIRM button: LED color: LED2, LED1, LED0, BUZ: **OFF**, LED: **OFF**, SEG: **ON**

[5] CONFIRM button, Fn key : Fn key decrement mode: **Not used**, CONFIRM button: **Used**, Fn key: **Used**

[6] Flashing speed: display status after pressing the CONFIRM button, display status, status after pressing the Fn key: **Standard**

[7] Display status after pressing the Fn key: BUZ, LED color, LED: **No change**, SEG: **OFF**

Note

(1) For “Repeatable,” do not go over the maximum command length when specifying.

(2) When setting up the data parameter for [3] ~ [7], make sure they do not result in “**02h”** or “**03h**” for **Ethernet** and **PCI** controllers, and “**0Dh**’ for the **RS-232C** controllers.

(3) In the case of **XW,** always send the “**M**” command before sending the operation instruction command.

Am1 (Display Status Specification)

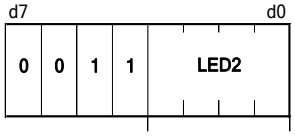
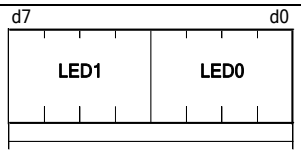
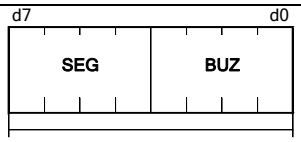
TW | TW-P | Ethernet | PCI | RS-232C | Repeatable | Wildcard

Function

Specifies the display status to each light module.

Format

A	m	1	[1] Address	[2]	[3]	[4]

No.	Items	Details
[1]	Address	Light Module Address
[2]	Display Status: LED2	 **0001**: OFF **0010**: ON **0011**: Flashing **0100**: High speed Flashing **1111**: No change
[3]	Display Status: LED1, LED0	 **0001**: OFF **0010**: ON **0011**: Flashing **0100**: High speed flashing **1111**: No change
[4]	Display Status: SEG, BUZ	 **0001**: OFF **0010**: ON **0011**: Flash **0100**: High speed flashing **1111**: No change

Maximum Command Length

Ethernet Type	64 Addresses
Board Type	
RS-232C Type	

Command Example

A	m	1	[1] Address				[2]	[3]	[4]
A	m	1	0	0	0	3	32h	22h	23h

Explanation

[1] Light module address: **0003**
[2] LED2: **ON**
[3] LED1, LED0: **ON**
[4] SEG: ON, BUZ: **Flash**

Note

(1) For “Repeatable,” do not exceed the maximum command length.

Am2 (Display Status Specification after Pressing the CONFIRM button)

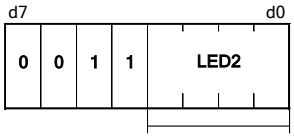
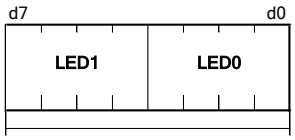
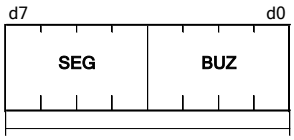
TW | TW-P | Ethernet | PCI | RS-232C | Repeatable | Wildcard

Function

Specifies the display status after pressing the **CONFIRM** button

Format

A	m	2	[1] Address	[2]	[3]	[4]

No.	Items	Details
[1]	Address	Light Module Address
[2]	Display Status: LED2	 **0001**: OFF **0011**: Flash **0010**: ON **0100**: High speed flashing **1111**: No change
[3]	Display Status: LED1, LED0	 **0001**: OFF **0011**: Flash **0010**: ON **0100**: High speed flashing **1111**: No change
[4]	Display Status: SEG, BUZ	 **0001**: OFF **0011**: Flash **0010**: ON **0100**: High speed flashing **1111**: No change

Maximum Command Length

Ethernet Type	64 Addresses
Board Type	
RS-232CC Type	

Command Example

A	m	2	[1] Address				[2]	[3]	[4]
A	m	2	0	5	2	4	31h	11h	21h

Explanation

[1] Light module address: **0524**

[2] LED2: **OFF**

[3] LED1, LED0: **OFF**

[4] SEG: ON, BUZ: **OFF**

Note

To do “Repeatable”, do not exceed the maximum command length.

Am3 (Display status specification after pressing the Fn key)

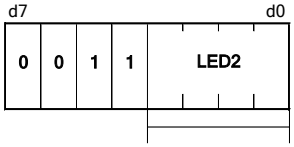
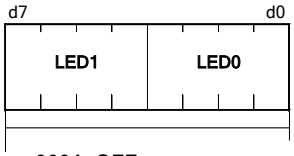
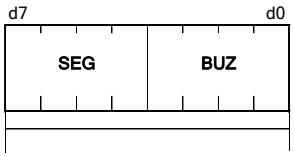
TW | TW-P | Ethernet | PCI | RS-232C | Repeatable | Wildcard

Function

Specifies the display status after pressing the **Fn** key.

Format



No.	Items	Details
[1]	Address	Light Module Address
[2]	Display Status: LED2	 **0001**: OFF **0011**: Flash **0010**: ON **0100**: High speed flashing **1111**: No change
[3]	Display status: LED1, LED0	 **0001**: OFF **0011**: Flash **0010**: ON **0100**: High speed flashing **1111**: No change
[4]	Display status: SEG, BUZ	 **0001**: OFF **0011**: Flash **0010**: ON **0100**: High speed flashing **1111**: No change

Maximum Command Length

Ethernet Type	64 Addresses
Board Type	
RS-232C Type	

Command Example

A	m	3	[1] Address				[2]	[3]	[4]
A	m	3	0	0	1	5	33h	33h	31h

Explanation

[1] Light module address: **0015**
[2] LED2: **Flash**
[3] LED1, LED0: **Flash**
[4] SEG: **Flash**, BUZ: **OFF**

Note

When doing “Repeatable,” do not exceed the maximum command length.

Am4 (CONFIRM Button and Fn Key Specification)

TW | TW-P | Ethernet | PCI | RS-232C | Repeatable | Wildcard

Function

Specifies the use / not use of the **CONFIRM** button and **Fn** key

Format

A	m	4	[1] Address	[2]

No.	Item	Details	Note
[1]	Address	Light module address	
[2]	Specify the **CONFIRM** button and **Fn** key	d7 d0 0 1 [a] [b] [c] **[a] Fn key decrement mode** **00**: Not use **01**: Use **11**: No change **[b] Fn key** **00**: Use **01**: Not use **11**: No change **[d] CONFIRM button** **00**: Use **01**: Not use **11**: No change	When the **Fn** decrement key is pressed during operation, the light module will display the quantity decreased by 1. When the key is pressed at 0, it will go back to the original. When the **CONFIRM** button is pressed, the quantity is captured by the application.

Maximum Command Length

Ethernet Type	64 Addresses
Board Type	
RS-232C Type	

Command Example

A	m	4	[1] Address	[2]
A	m	4	2 4 0 0	50h

Explanation

[1] Light module address: **2400**

[2] CONFIRM button, Fn key specification: Fn key decrement mode: **Use**, CONFIRM button: **Use**

Note

When using “Repeatable,” do not exceed the maximum command length.

Am7 (Display Data Specification after Pressing the Fn key)

TW | TW-P | Ethernet | PCI | RS-232C | Repeatable | Wildcard

Function

Specifies the display data after pressing the **Fn** key.

Format

A	m	7	[1] Address	[2] Display Data

No.	Item	Details	Note
[1]	Address	Light module address	
[2]	Display data	Display data	Always specify 5 characters

Maximum Command Length

Ethernet Type	64 addresses
Board Type	
RS-232C Type	

Command Example

A	m	7	[1] Address				[2] Display Data				
A	m	7	0	2	0	0	9	9	9	9	9

Explanation

[1] Light module address: **0200**

[2] Display data: **99999**

Note

(1) This command is sent before sending the operation specification command.

(2) The default is “-----”.

(3) Do not exceed the maximum command length when doing “repeatable.”

FAm7 (Display Data Specification After Pressing the Fn Key -- TW2060 Only)

TW | TW-P | Ethernet | PCI | RS-232C | Repeatable | Wildcard

Function

Specifies the display data after pressing the **Fn** key of the TW2060.

Format

F	A	m	7	[1]	[2] Address	[3] Display Data After Pressing Fn

No.	Item	Details	Note
[1]	[3] The number of characters displayed after pressing the **Fn** key	Possible specification range:01~20	Outside the range of 01~20, the controller will send a negative response "**n**".
[2]	Address	Light module address	Possible specification range is 1~7999.
[3]	Display data after pressing the **Fn** key	The display data after **Fn** key is pressed. The length varies.	Always fill the number of characters specified by [1].

Maximum Command Length

Ethernet type	1015 bytes
Board type	1024 bytes
RS-232C type	

Command Example

F	A	m	7	[1]		[2] Address				[3] Display Data After Pressing Fn									
F	A	m	7	1	1	1	2	0	1	S	H	O	P		C	O	D	E	9	9

Explanation

[1] Number of characters of the display data after pressing the Fn key: **11**

[2] Light module address: **1201**

[3] Display data after pressing Fn key: **SHOP CODE99**

Note

(1) This command is sent before sending the operation instruction command.

(2) Default is "------".

(3) If the displayed data exceeds 6 characters after pressing the **Fn** key, use it with the "**Amb**" command (the scroll status specification after pressing the **Fn** key).

(4) For repeatable, do not exceed the maximum command length.

Am8 (Poka Yoke Mode Specification) TW2916 Exclusive

TW | TW-P | Ethernet | PCI | RS-232C | Repeatable | Wildcard

Function

Specifies various functions for “poka yoke” (error prevention) light modules.

Format

A	m	8	[1] Address	[2]	[3]

No.	Item	Details	Note
[1]	Address	Light module address	
[2]	Automatic function specification	d7 d0 0 1 [a] [b] [c] **[a] “Poka Yoke” function** **00**: Use ∗ **01**: Not use **11**: No change **[b] Auto CONFIRM** **00**: Use ∗ **01**: Not use **11**: No change **[d] Auto CONFIRM timing** **00**: Hand retracts from the location ∗ **01**: Hand enters the location **11**: No change	What is “Poka Yoke” action? LED1 [green] turns OFF when a hand enters the location. LED 2 [red] flashes at a high speed and BUZ turns ON. ∗Default
[3]	Sensor reaction time specification	d7 d0 0 1 [a] [b] **[a] Sensor reaction time when a hand retracts from a location** **000**: 0 sec ∗ **001**: 1 sec **010**: 2 sec **011:** 3 sec **111:** No change **[b] Sensor reaction time when a hand enters a location** **000**: 0 sec ∗ **001**: 0,1 sec **010**: 0.2 sec **010**: 0.5 sec **111**: No change	∗Default

Maximum Command Length

Ethernet type	64 addresses
Board type	
RS-232C type	

Command Example

A	m	8	[1] Address				[2]	[3]
A	m	8	2	4	0	0	47h	7Fh

Explanation

[1] “Poke yoke” light module address: **2400**

[2] Automatic function specification: “poka yoke” function: **Use**, Automatic CONFIRM button: **Not use**
(Operation：When detected by the sensor, a warning by a buzzer)

[3] Sensor reaction time specification: Sensor reaction time when a hand is retracted from the location: **No change**, When a hand goes into the location: **No change**

Sensor Reaction Time Specification

■What does it mean by the sensor reaction time when a hand enters a location?

The time it takes to detect when an operator’s hand goes into a location.

When the sensor reaction time is 0 sec, the moment an operator’s hand enters a location ever for an instant, the “poka yoke” light module determines that a hand is in the location and performs an “automatic function specification” operation. Certain specified sensor reaction time, however, is not desirable. For example, suppose this reaction time to perform the automatic “poka yoke” action is 0.5 sec. From the moment an operator’s hand goes into and removed the location is within 0.5 sec, the buzzer will not sound a warning. Instead it will start after 0.5 sec.

■What does it mean by the sensor reaction time when a hand is retracted from a location?

It is the time it takes to detect when an operator’s hand is completely retracted from a location. If this reaction time is 0 sec and an operator’s hand is retracted even for an instant after entering a location, the “poka yoke” light module will determine that the hand was pulled away and will perform the specified “automatic function” operation. Certain specified sensor reaction time will not work.

In the above 2 examples, 2 sensor reaction setup times are given; however, try selecting an appropriate value by testing at the actual site.

Note

(1) Send the “**Am8**” command before sending the operation instruction command.

(2) Do not exceed the maximum command length when doing Repeatable.

Ama (Specifies the Use of DigNum Keys)

TW | TW-P | Ethernet | PCI | RS-232C | Repeatable | Wildcard

Function

Specify the use of the TW2060's quantity revision keys

Format

A	m	a	[1] Address	[2]

No.	Item	Details	Note
[1]	Address	Light module address	Possible specification range: 1~7999
[2]	Specify the quantity revision key	d7 d0 0 1 0 0 [a] [b] [a] Quantity revision mode 00: Normal (Space is not treated as a number value) 01: Special (Space is treated as a number value) [b] Quantity revision key 00: DigNum key 01: Not used 10: +/- key 11: No change	**DigNum**: Corresponds to the "**S**" SET key, "**-**" minus key. **Special** mode is effective only when the **+/-** key is used for the quantity revision.

Maximum Command Length

Ethernet type	64 addresses
Board type	
RS-232C type	

Command Example

A	m	a	[1] Address	[2]
A	m	a	0 1 2 0	41h

Explanation

[1] Light module address**: 0120**

[2] Quantity revision key use: **Not used**

Note

(1) The 2060 cannot use the **+/-** key to revise quantity.

(2) Do not exceed the maximum command length to do "Repeatable."

Amb (Scroll Status Specification After Pressing the Fn key) TW2060 Exclusive

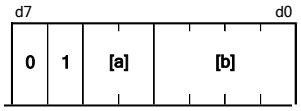
TW | TW-P | Ethernet | PCI | RS-232C | Repeatable | Wildcard

Function

Specifies the scrolling status of the TW2060 type module after pressing the **Fn** key.

Format

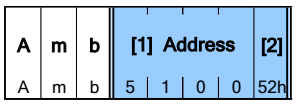


No.	Item	Detail	Note
[1]	Address	Light module	Possible specification range: 1~7999
[2]	Scrolling mode, scrolling speed setup	 **[a] Scrolling mode** **00**: No scrolling **01**: Scrolling ON **10**: Scrolling ON (with pause) **11:** No change **[b] Scrolling speed** **0001**: Normal **0010**: Fast **0011**: Fastest **1111**: No change	

Maximum Command Length

Ethernet type	64 addresses
Board type	
RS-232Ctype	

Command Example



Explanation

［1］Light module address: **5100**

［2］Scrolling mode: Scroll **ON**, Scrolling speed: **Fast** (high speed)

Note

For “Repeatable”, do not exceed the maximum command length.

Amd (Expanded Operation Mode Specification) TW2414 Exclusive

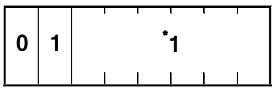
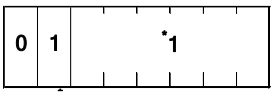
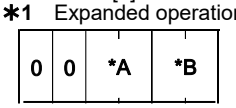
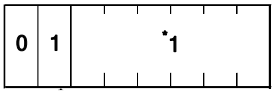
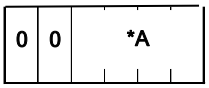
TW | TW-P | Ethernet | PCI | RS-232C | Repeatable | Wildcard

Function

Set up for expanding the module function.

Format



No.	Item	Detail
[1]	Address	Module address
[2]	Operation number	 *1 Operation number 000001 TW2414 expanded operation
[3]	Expanded operation mode	 When *1 in [2] is 000001: TW2414 expansion operation ✱1 Expanded operation mode  ✱A A pause after Q2, X process 00: Not use 01: Use ✱B Left justified 00: Not use 01: Use
[4]	Expanded operation mode	 When *1 in [2] is 000001: TW2414 expansion operation ✱1 Expanded operation mode  ✱A Specifies the number of 14 SEG on module 0100: 4 1001: 9 0101 5 1010: 10 0110 6 1011: 11 0111: 7 1100 12 1000: 8

Command Example

A	m	d	[1] Address				[2]	[3]	[4]
A	m	d	8	1	0	1	41h	45h	4Ah

Explanation
[1] Module address: **8101**
[2] TW2414 Expanded operation:
[3] **Scroll with pause**, **Left** justified
[4] Number of 14SEG: **10**

Note

The module that can use the operation mode=1 (000001) is TW2414 Ver.1.5 or higher (hereafter on it will be called TW2414).

Expanded Operation Mode Explanation and Restrictions on Operation Number =1

■Expanded Operation Mode

[***A** Scroll and pause after the "**Q2**", "**X**" processing]

When a display code is sent with the "**Q2**" or "**X**" command using the TW2414 with the "**01: use**" setup with the "**Amd**" command and "**pause and scroll**" with the "**Am1**" command, there will be a pause at the beginning followed by scroll and display. This makes it easy for an operator to recognize words.

When the setup is "**00: not use**", the display begins with scrolling.

[***B** Left Justified Display]

When the "**Amd**" command setting is **"01: use**", the display will be left justified regardless of the number of alphanumeric characters sent by the "**Q2**" command. If the setting is "**00: not use**", the characters will be left justified from the 12th character. If less than 12 characters, they will be right justified.

■ Expanded Operation Mode

[***A** Setting up the number of 14SEG]

Determine how many alphanumeric characters to display. The number of 14SEG can be stored in the TW2414 with "**Amd**" function. Previously, when more than 10 characters (not including 10) are specified, the first 2 characters were truncated. But the "**Q2**" and "**X**" commands eliminated this limitation. Usually specify "**1010:10**"

■Limitations

If the "**Amd**" command is used when a character string is display, the display will be clear. Resend the "**Q2**" and "**X**" commands to correct the display.

Ame (Condition Specification for Each SEG Display)

Ethernet | Repeatable | Wildcard

Function

Specifies operation condition of each SEG.

Format

A	m	e	[1] Address	[2]	[3]

No.	Item	Detail	Note
[1]	Address	Light module address	Possible specification range: 1~7999
[2]	Applied timing	**1**: When an operation instruction is given / Reverse display **2**: After pressing the **CONFIRM** button / Reverse display **3**: After pressing the **Fn** key / Reverse display **4**: When an operation instruction is given / Small font display **5**: After pressing the **CONFIRM** button / Small font display **6**: After pressing the **Fn** key / Small font display	
[3]	Display mode of each digit	d7 … d0 0 1 0 [a] [b] [c] [d] [e] **[a]** 5th digit, **[b]** 4th digit, **[c]** 3rd digit **[d]** 2nd digit, **[e]** 1st digit **0**: Normal operation **1**: Applied timing operation	

Maximum Command Length

Ethernet type	64 addresses
Board type	
RS-232C type	

Command Example

A	m	e	[1] Address				[2]	[3]
A	m	e	0	1	0	5	2	44h

Explanation

[1] Light module address: **0105**

[2] Applied timing: **After pressing the CONFIRM button**

[3 Display mode of each digit: **The display mode of the 3rd digit is reversed.**

Note

None.

4.5. Operation Instruction Commands

When the controller receives the operation command from the PC, the controller sends the operation instruction to the light module and monitors the progress.
Monitoring the operation progress or status means the controller continues to monitor the progress until the light modules that received the instruction either have a normal completion or a shortage report. In addition, it considers every signal light or batch display as 1 block. When the operation or shortage report from all the light modules within that block is received, the signal light or batch display of the block will be turned OFF. The operation status report, such as which light module has completed the operation, etc., is stored after the completion report is sent to the PC.

Operation Instruction Commands At-a-glance

Command	Setup details	Applicable		Page No.
		TW	XW	
PP1	Block operation instruction with display data specified after pressing the **Fn** key	○		51
PP5	Rewritable block operation instruction with display data specified after pressing the **Fn** key	○		52
P1	Block operation instruction	○	○	55
P4	Block operation instruction with operation mode	○	○	56
P5	Rewritable block operation instruction with operation mode	○	○	58
L	Individual operation instruction.	○		60
FP1	Block operation instruction with display data specified after pressing the **Fn** key for TW2060 type.	○		61
FP5	Rewritable block operation instruction with display data specified after pressing the **Fn** key for TW2060 type.	○		63

○: Applicable

4.5.1 +/- key, Fn key Decrement Mode

When sending an operation instruction to the light module by using the **+/-** key or **Fn** key decrement mode to revise quantity, put a space for the digit column that does not require revision. For example, the instruction for “ 1” (4 spaces and 1), 0 ~ 9 can be processed with the **+/- key;** if the instruction says “ 01” (3 spaces and 01), 00 ~ 99 can be processed.

4.5.2 Block No.

When the last 2 digits of the signal light address are specified (“00” for the signal light address 8000), the controller turns the signal light ON/OFF automatically. However, the number of light modules that can be specified within a block is limited to 255.
If the last 2 digits of the batch display or signal light match the 2 digits of the block number, the controller will automatically turn ON/OFF the device.

The following applies to PCI board type and the RS-232C type controllers

Even when the automatic ON/OFF function is not used or non-existent, always set up 00~99 in binary-coded decimal (BCD). Light modules can be grouped using the application. However, using more than 10 Block Numbers can cause a time delay when the specification completion check (the **Hxx** command) is used. Checking the completion of 1 Block No. is 1/N times, (block by block) may cause a delay in checking. Find ways to group the operation appropriately. If grouping is not necessary or takes too much time, we recommend specifying the same Block No.

4.5.3 Note

Wildcards cannot be used for the addresses in the operation specification command.

PP1 (Block Operation Instruction with Display Data specified after Pressing the Fn key)

TW | TW-P | Ethernet | PCI | RS-232C | Repeatable | MA

Function

The block operation instruction that specifies the display data and the display data after pressing the Fn key. (If Mode Array is specified, send the operation mode to the light modules)

Format

(1) Operation mode specification・Operation instruction

P	P	1	0	5	[1]	[2] Block No.	[3] Mode Array	[4] Address	[5] Display Data

(3) Operation mode specification ・Operation instruction ・specify display data after pressing the Fn key

P	P	1	0	5	[1]	[2] Block No.	[3] Mode Array	[4] Address	[5] Display Data	[6] Display Data After Pressing Fn

No.	Item	Detail	Note
[1]	Displayed data specification after pressing the **Fn** key	00 : Not specified 05 : Specified	Values that can be set up – **only "00" or "05"** 00 : [6] omitted 05 : Always specify 5 characters for [6]
[2]	Block No.	Block number	Possible specification range: 00~99
[3]	Mode Array	Refer to [mode array] Variable length. Can be omitted	When mode array is omitted, continue from [4] after [2].
[4]	Address	Light module address	Possible range of specification: 0001~7999
[5]	Display Data	Displayed data	Always specify 5 characters
[6]	Display Data After Pressing the **Fn** key	Displayed data after pressing the **Fn** key	Omitted when [1] is 00 Always specify 5 characters only when [1] is 05

Maximum Command Length

1015 bytes

Command Example

(1) Operation mode specification • Operation instruction • Specify display data after pressing the **Fn** key

P	P	1	0	5	[1]		[2] Block No.		[3]Mode Array								[4] Address				[5] Display Data				[6]Display Data After Pressing Fn				
P	P	1	0	5	0	5	0	1	m	1	34h	44h	FFh	m	3	3Fh	FFh	3Fh	1	0	0	1	20h	20h	20h	20h	3	9	8	7	6	5

Explanation

[1] Display data after pressing the **Fn** key: **Specified**

[2] Block Number: **01**

[3] Display status, Display status after pressing the **Fn** key: **Specified**

[4] Light module address: **1001**

[5] Display data: “ **3**” (space, space, space, space 3)

[6] Display data after pressing the **Fn** key**:** “**98765**”

(2) Operation mode specification • Operation instruction

P	P	1	0	5	[1]		[2] Block No.		[3]Mode Array				[4] Address				[5] Display Data				
P	P	1	0	5	0	0	0	1	m	2	32h	22h	21h	1	0	0	1	20h	20h	20h	20h	5

Explanation

[1] Display data after pressing the **Fn** key**: Not specified**

[2] Block number: **01**

[3] Display status after pressing the **CONFIRM** button: **Specified**

[4] Light module address: **1001**

[5] Display data: “ **5**” (space, space, space, space, 5)

(3) Repeatable, operation instruction, and display data specification after pressing the **Fn** key.

P	P	1	0	5	[1]		[2] Block No.		…
P	P	1	0	5	0	5	0	1	

…	[4] Address				[5] Display Data					[6] Display Data After Pressing Fn					[4] Address				[5] Display Data					[6] Display Data After Pressing Fn				
	1	0	0	1	0	0	0	1	5	1	-	2	3	4	1	0	0	2	0	0	0	0	1	9	-	8	7	6
	Address 1001														Address 1002													

Explanation

[1] Display data after pressing the **Fn** key: **Specified**

[2] Block number: **01**

[3] Mode array: **Omitted**

[4] Light module address: **1001, 1002**

[5] Display data: “**00015**”**,** “**00001**”

[6] Display data after pressing the **Fn** key: “**1-234**”, “**9-876**”

Operation

Multiple operation instructions are sent to the block (signal light or batch display) simultaneously.

Note

When doing “Repeatable,” do not exceed the maximum command length.

PP5 (Rewritable Block Operation Instruction with Display Data Specified after Pressing the Fn Key)

TW | TW-P | Ethernet | PCI | RS-232C | Repeatable | MA

Function

Specify the display data and the display data after pressing the **Fn** key while giving the rewritable block operation instruction to the light modules. (If a Mode Array is specified, send the operation mode to the light modules.)

Format

(1) Specify the operation Mode • Give an operation instruction

P	P	5	0	5	[1]	[2] Block No.	[3] Mode Array	[4] Address	[5] Display Data

(2) Specify the operation mode • Give an operation instruction • Specify display data after pressing the Fn key

P	P	5	0	5	[1]	[2] Block No.	[3] Mode Array	[4] Address	[5] Display Data	[6] Display Data After Pressing Fn

No.	Item	Details	Note
[1]	Display data specification after pressing the **Fn** key	00 : Not specified 05 : Specified	Values that can be specified: Only **00** or **05** 00 : [6] is omitted 05 : Always specify 5 characters for [6]
[2]	Block No.	Block number	Possible specification range is 00 ~ 99.
[3]	Mode Array	Refer to [Mode Array]. Variable length can omit	When omitting the “Mode Array”, continue from [4] after [2].
[4]	Address	Light module address	Can specify 0001 ~ 7999.
[5]	Display Data	Display data	Always specify 5 characters
[6]	Display Data After Pressing the **Fn** key	Display data after pressing the **Fn** key	When [1] is 00, omit. Only when [1] is 05, always specify 5 characters.

Maximum Command Length

1015 bytes

Command Example

(1) Specify the operation mode • Give operation instruction

P	P	5	0	5	[1]		[2] Block No.		[3] Mode Array					[4] Address				[5] Display Data				
P	P	5	0	5	0	0	9	9	m	2	31h	11h	21h	1	2	0	1	20h	20h	20h	1	5

Explanation

[1] Display data after pressing the Fn key: **Not specified**
[2] Block number: **99**
[3] Display status after pressing the CONFIRM button: **Specified**
[4] Light module address: **1201**
[5] Display data: **“ 15”** (space, space, space, 15)

(4) Specify the operation mode • Give operation instruction • Specify the display data after pressing the **Fn** key

P	P	5	0	5	[1]		[2] Block No.		[3] Mode Array									[4] Address				[5] Display Data					[6] Display Data After Pressing Fn				
P	P	5	0	5	0	5	9	9	m	1	31h	31h	21h	m	3	3Fh	FFh	3Fh	0	1	0	1	0	0	0	0	5	6	6	6	6	6

Explanation

[1] Display data after pressing the Fn key: **Specified**
[2] Block number: **99**
[3] Display status, Display status after pressing the Fn key: **Specified**
[4] Light module address: **0101**
[5] Display data: “**00005**”
[6] Display data after pressing the Fn key: “**66666**”

(5) Omit the operation mode specification • Give the operation instruction

P	P	5	0	5	[1]		[2] Block No.		[4] Address				[5] Display Data				
P	P	5	0	5	0	0	0	0	0	0	0	1	0	0	0	1	2

Explanation

[1] Displayed data after pressing the Fn key: **Not specified**
[2] Block number: **00**
[3] Mode array: **Omitted**
[4] Light module address: **0001**
[5] Display data: “**00012**”

Operation

Several operation instructions are sent to the block unit (signal light or batch display) all at once, and they can be sent directly to the light modules that are in operation (rewrite).

Note

When doing “Repeatable,” do not exceed the maximum command length.

P1 (Block Operation Instruction)

TW | TW-P | XW | Ethernet | PCI | RS-232C | Repeatable

Function

Gives the block operation instruction

Format

Operation instruction



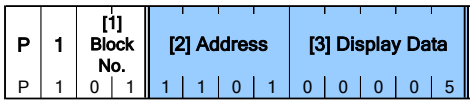
No.	Item	Details	Note
[1]	Block No.	Block number	Specification range: 00~99
[2]	Address	Light module address	Specification range: 0001~7999
[3]	Display Data	Display data	Must specify 5 characters

Maximum Command Length

Ethernet type	1015 bytes
Board type	128 addresses
RS-232C type	

Command Example

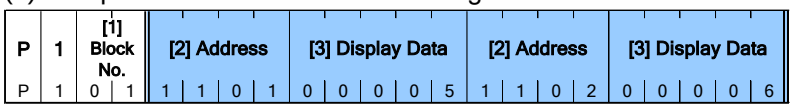
(1) Operation instruction



Explanation

[1] Block number: **01**
[2] Light module address: **1101**
[3] Display data: "**00005**"

(2) Operation instruction to several light modules



Explanation

[1] Block number**: 01**
[2] Light module addresses: **1101,1102**
[3] Display data: "**00005**", "**00006**"

Operation

Several operation instructions are sent to the block unit (signal light or batch display) all at once.

Note

When doing "Repeatable", do not exceed the maximum command length.

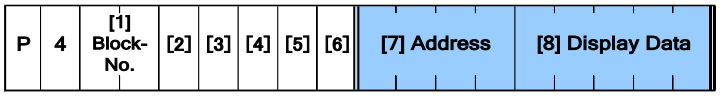
P4 (Block Operation Instruction with Operation Mode)

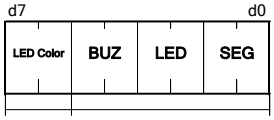
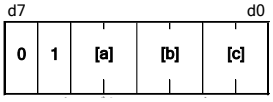
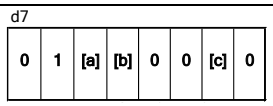
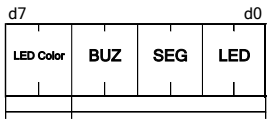
TW | TW-P | XW | Ethernet | PCI | RS-232C | Repeatable

Function

Indicates block operations of light modules with the specified operation mode

Format



No.	Name	Details	Note
[1]	Block No.	Block Number	Specification range: 00 ~ 99
[2]	Light module status	 **00**: LED 2, 1, 0 **00**: OFF **01**: LED 2, 0 **01**: ON **10**: LED 1 **10**: Flash **11**: LED 2, 0 **11**: No change	XW LED color is fixed at 01 TW If FFh, no change
[3]	Display status after pressing the **CONFIRM** button		
[4]	**CONFIRM** button, **Fn** key	 (a) **Fn** key decrement mode **00**: Not use **01**: Use **11**: No change (b) **Fn** key **00**: Use **01**: Not use **11**: No change (c) **CONFIRM** button **00**: Use **01**: Not use **11**: No change	XW (a) Fn key decrement mode is fixed at 00 TW If FFh, no change
[5]	Flashing or blinking speed	 [a] Display status after the **CONFIRM** button is pressed [b] Display status 0: Normal [c] Status after the **Fn** Key is pressed 1: High speed	Flashing speed when flashed with the condition [2], [3], [6]. XW [a] After pressing the **CONFIRM** button, the display is fixed at 0. [c] After pressing the **Fn** key, the display status is fixed at 0.
[6]	Display data after pressing the **Fn** key	 **00**: LED 2, 1, 0 **00**: OFF **01**: LED 2, 0 **01**: ON **10**: LED 1 **10**: Flash **11**: LED 2, 0 **11**: No change	XW LED color is fixed at 01 BUZ and SEG are fixed at 00 For TW If FFh, no change
[7]	Address	Light module address	Specification range: 0001 ~ 7999
[8]	Display Data	Displayed data	Always specify 5 characters

Maximum Command Length

Ethernet type	1015 bytes
Board type	128 addresses
RS-232C type	

Command Example

P	4	[1] Block-No.		[2]	[3]	[4]	[5]	[6]	[7] Address				[8] Display Data				
P	4	1	0	09h	40h	41h	40h	40h	0	5	2	0	20h	20h	20h	20h	5

Explanation

[1] Block number: **10**

[2] Display status: LED color: **LED2, LED1, LED0**; BUZ: **OFF**; LED: **Flashing**; SEG: **ON**

[3] Display status after pressing the CONFIRM button: LED color: **LED1**; BUZ: **OFF**; LED: **OFF**; SEG: **OFF**

[4] CONFIRM button, Fn key: Fn key decrement mode: **Not used**; CONFIRM button: **Not used**, Fn key: **Use**;

[5] Flashing or blinking speed: Display status after pressing the CONFIRM button and the Fn key, Display status: **Normal**

[6] Display status after pressing the Fn key: LED color: **LED1**; BUZ: **OFF**; LED: **OFF**; SEG: **OFF**.

[7] Light module address: **0520**

[8] Display data: “ **5**” (space, space, space, space, 5)

Operation

Several operation instructions are sent to the block unit (signal light or batch display) simultaneously.

Note

When doing “Repeatable,” do not exceed the maximum command length.

P5 (Rewritable Block Operation Instruction with Operation Mode)

TW | TW-P | XW | Ethernet | PCI | RS-232C | Repeatable

Function

Instruct block operation with light module mode specification, perform rewritable.

Format

P	5	[1] Block-No.	[2]	[3]	[4]	[5]	[6]	[7] Address	[8] Display Data

No.	Item	Details	Note
[1]	Block No.	Block Number	Range specification: 00 ~ 99
[2]	Display status	d7 d0: LED Color / BUZ / LED / SEG **00**: LED 2, 1, 0 **00**: OFF **01**: LED 2, 0 **01**: ON **10**: LED 1 **10**: Flash **11**: LED 2, 0 **11**: No change	XW LED color is fixed at 01 TW No change if **FFh**
[3]	Status after pressing the **CONFIRM** button		
[4]	**CONFIRM** button, **Fn** Key	d7 d0: 0 / 1 / [a] / [b] / [c] (c) **Fn** key decrement mode: **00**: Not use **01**: Use **11**: No change (d) **Fn** key: **00**: Use **01**: Not use **11**: No change (c) **CONFIRM** button: **00**: Use **01**: Not use **11**: No change	XW (a) The **Fn** key decrement mode is fixed at 00. TW No change if FFh
[5]	Flashing or blinking speed	d7: 0 / 1 / [a] / [b] / 0 / 0 / [c] / 0 [a] Displayed status after the **CONFIRM** button press [b] Displayed status [c] Status after the **Fn** Key press 0: Normal 1: High speed	The flashing speed when flashed with the condition [2], [3], [6]. XW [a] The display status after pressing the **CONFIRM** button is fixed at 0. [c] The display status after pressing the **Fn** key is fixed at 0.
[6]	Status after pressing the **Fn** key	d7 d0: LED Color / BUZ / SEG / LED **00**: LED 2, 1, 0 **00**: OFF **01**: LED 2, 0 **01**: ON **10**: LED 1 **10**: Flash **11**: LED 2, 0 **11**: No change	XW LED color is fixed at 01. BUZ and SEG are fixed at 00. TW No change if FFh
[7]	Address	Light module address	Specification range:0001 ~ 7999
[8]	Display Data	Display data	Always specify 5 characters

Maximum Command Length

Ethernet type	1015 bytes
Board type	128 addresses
RS-232C type	

Command Example

P	5	[1] Block-No.		[2]	[3]	[4]	[5]	[6]	[7] Address				[8] Display Data				
P	5	1	0	09h	40h	41h	40h	40h	0	5	2	0	20h	20h	20h	20h	5

Explanation

[1] Block number: **10**

[2] Display status: LED Color: **LED2, LED1, LED0**; BUZ: **OFF**; LED: **Flash**; SEG: **ON**

[3] Status after pressing the CONFIRM button: LED color: **LED1**; BUZ: **OFF**; LED: **OFF**; SEG: **OFF**

[4] CONFIRM button, Fn Key: Fn key decrement mode: **Not use**; CONFIRM button: **Not use**; Fn key: **Use**

[5] Flashing speed: Display status after pressing the CONFIRM button, Display status, Display status after pressing the Fn key: **Standard**

[6] Display status after pressing the Fn key: LED color: **LED1**; BUZ: **OFF**; LED: **OFF**; SEG: **OFF**

[7] Light module address: **0520**

[8] Display data: “ **5**” (space, space, space, space, 5)

Operation

Several operation instructions are sent to the block unit (signal light or batch display) simultaneously, and they can be sent directly to the light modules that are in operation. (rewrite)

Note

When doing “Repeatable,” do not exceed the maximum command length.

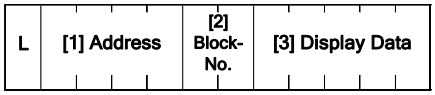
L (Individual Operation Instruction)

TW | TW-P | XW | Ethernet | PCI | RS-232C

Function

Performs individual operation instruction

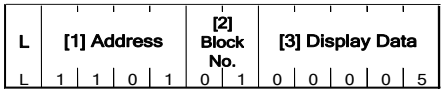
Format



No.	Item	Details	Note
[1]	Address	Light module address	Specification range: 0001 ~ 7999
[2]	Block No.	Block number	Specification range: 00 ~ 99
[3]	Display Data	Display data	Always specify 5 characters

Command Example

Operation Instruction



Explanation

[1] Light module address: **1101**
[2] Block number: **01**
[3] Display data: "**00005**"

Operation

Send the operation instruction to the specified address.

FP1 (Block Operation Instruction with Display Data Specified after Pressing the Fn Key) for TW2060

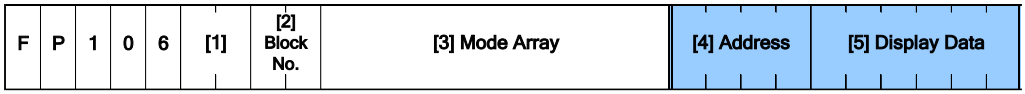
TW | TW-P | Ethernet | PCI | RS-232C | Repeatable | MA

Function

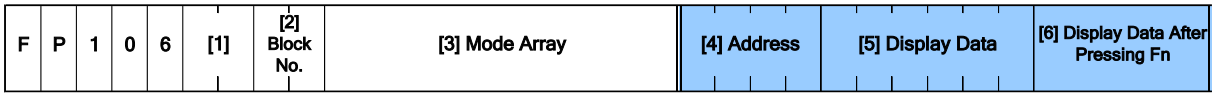
Give block operation instruction to light modules with specification of display data, and the display data and data length after pressing the **Fn** key.

Format

(1) Operation mode specification • Operation instruction



(2) Operation mode specification • Operation instruction • Display data specification after pressing the **Fn** key



No.	Item	Detail	Note
[1]	Specify display data after pressing the **Fn** key	00: Not specified 01 ~ 20: Number of specified characters	00: [6] is omitted
[2]	Block No.	Block number	Specification range: 01 ~ 99
[3]	Mode Array	See "**Mode Array**" Variable length; Can be omitted.	When Mode Array is omitted, continue from [Address].
[4]	Address	Light module address	Specification range: 0001 ~7999
[5]	Display Data	Display data	Always specify 6 characters.
[6]	Display Data After Pressing **Fn**	Display data after pressing the **Fn** key	Specify the number of characters specified in [1].

Maximum Command Length

1015 bytes

Command Example

(1) Operation Mode Specification • Operation Instruction • Display data specification after pressing the Fn key

F	P	1	0	6	[1]		[2] Block No.		[3] Mode Array			[4] Address				[5] Display Data						[6] Display Data After Pressing Fn									
F	P	1	0	6	1	1	0	1	m	b	52h	1	0	0	1	1	2	-	0	1	6	P	A	R	T	S		A	-	0	0	1

Explanation

[1] Number of display characters after pressing the Fn key: **11**
[2] Block number: **01**
[3] Scroll mode after pressing the Fn key: **High speed**
[4[Light module address: **1001**
[5] Display data: **12-016**
[6] Display data after pressing the Fn key: **PARTS A-001**

(2) Operation mode specification • Display data omitted after pressing the **Fn** key • Operation instruction

Explanation

F	P	1	0	6	[1]		[2] Block No.		[3] Mode Array				[4] Address				[5] Display Data					
F	P	1	0	6	0	0	0	1	m	2	31h	11h	21h	1	5	0	1	B	2	-	0	1	3

[1] Number of display characters after pressing the Fn key: **Omitted**
[2] Block number: **00**
[3] Display status after pressing the CONFIRM button: **Specified**
[4] Light module address: **1501**
[5] Display data: **B2-013**
[6] Display data after pressing the Fn key: **Omitted**

Note

When doing “Repeatable,” do not exceed the maximum command length.

FP5 (Rewritable Block Operation Instruction with Display Data Specified after Pressing the Fn key) for TW2060

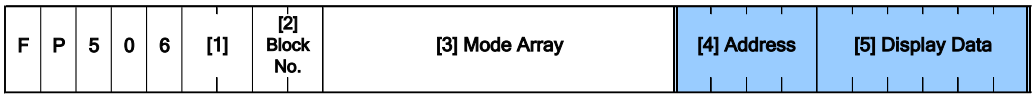
TW | TW-P | Ethernet | PCI | RS-232C | Repeatable | MA

Function

The "**FP5**" command is a rewritable block operation specification command with display data specified after pressing the **Fn** key. The data length and display data after pressing the **Fn** key can be specified.

Format

(1) Operation mode specification • Operation instruction



(2) Operation mode specification • Operation instruction • Display data specified after pressing the **Fn** key



No.	Item	Detail	Note
[1]	Specify the display data after pressing the **Fn** key	00: Not specified 01 ~ 20 : Specify number of characters	00: [6] is omitted
[2]	Block No.	Block number	Specification range: 01 ~99
[3]	Mode Array	Refer to "Mode Array" Variable length. Can be omitted.	When Mode Array is omitted, continue from "Address."
[4]	Address	Light module address	Specification range: 0001 ~ 7999
[5]	Display Data	Display data	Always specify 6 characters.
[6]	Display Data After Pressing **Fn**	Display data after pressing the **Fn** key	Specify with the number of characters specified in [1].

Maximum Command Length

1015 bytes

Command Example

(1) Operation mode specification • Operation instruction • Display data specified after pressing the Fn key

F	P	5	0	6	[1]		[2] Block No.		[3] Mode Array							[4] Address				[5] Display Data						[6] Display Data After Pressing Fn					
F	P	5	0	6	0	6	0	1	m	3	31h	21h	21h	m	b	52h	4	0	0	1	0	2	-	0	1	6	C	O	D	E	0	3

Explanation

[1] Number of displayed characters after pressing the Fn key: **6**
[2] Block number: **01**
[3] Display status after pressing the Fn key, Scroll mode after pressing the Fn key: **Specified**
[4] Light module address: **4001**
[5] Display data: **02-016**
[6] Display data after pressing the Fn key: **CODE03**

(2) Operation mode specification • Display data omitted after pressing the **Fn** key • Operation instruction

F	P	5	0	6	[1]		[2] Block No.		[3] Mode Array					[4] Address				[5] Display Data						
F	P	5	0	6	0	0	0	0	m	2	31h	21h	21h	0	1	0	1	C	A	-	0	1	6

Explanation

[1] Number of display characters after pressing the Fn key: **Omitted**
[2] Block number: **00**
[3] Display data after pressing the CONFIRM button: **Specified**
[4] Light module address: **0101**
[5] Display data: **CA-016**
[6] Display data after pressing the Fn key: **Omitted**

(3) Operation mode instruction omitted • Display data after pressing the **Fn** key omitted • Operation instruction given

F	P	5	0	6	[1]		[2] Block No.		[4] Address				[5] Display Data					
F	P	5	0	6	0	0	9	9	0	0	2	1	A	0	1	-	0	2

Explanation

[1] Number of display characters after the Fn key is pressed: **Omitted**
[2] Block number: **99**
[3] Operation mode specification: **Omitted**
[4] Light module address: **0021**
[5] Display data: **A01-02**
[6] Display data after pressing the Fn key: **Omitted**

Note

When doing “Repeatable,” do not exceed the maximum command length.

4.6. Special Page-Turning Commands

Special Page-Turning Command Chart

Command	Details	Applicable		Page No.
		TW	XW	
Pm	Page-turning operation instruction with mode array.	○		66
P6	Page-turning operation instruction with operation mode.	○		68

○ : Applicable

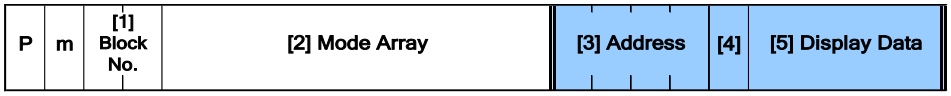
Pm (Page-turning Operation Instruction) with Mode Array

TW-P | Ethernet PCI | Repeatable MA

Function

Sends the page-turning operation instruction to each light module up to 10 pages.
If “Mode Array” is specified, the operation mode is sent to the light modules.

Format



No.	Item	Detail	Note
[1]	Block No.	Block number	Specification range: 00~99
[2]	Mode Array	Refer to “**Mode Array**” Variable length can omit	When Mode Array is omitted, continue from [3] after [1].
[3]	Address	Light module address	Specification range: 0001 ~ 7999
[4]	Page	Number of pages	Specification range: 0~9. 0: 1 page 9: 10 pages (max)
[5]	Display Data	Display data	Always specify 5 characters per page.

Maximum Command Length

Ethernet type	1015 bytes
Board type	1024 bytes
RS-232C type	

Command Example

(1) Operation mode specification • Page-turning operation instruction

P	m	[1] Block No.		[2] Mode Array					[3] Address				[4]	[5] Display Data (Page 1)					[5] Display Data (Page 2)					[5] Display Data (Page 3)				
P	m	0	1	m	1	32h	22h	FFh	0	0	0	1	2	1	-	20h	1	0	2	-	20h	2	0	3	-	20h	3	0

Explanation

[1] Block number: **01**
[2] Display status: **Specified**
[3] Light module address: **0001**
[4] Number of pages: **3**
[5] Display data: First page: "**1- 10**", 2nd page: "**2- 20**", 3rd page: "**3- 30**"

(2) Operation mode specification • Page-turning operation instruction (Repeatable)

P	m	[1] Block No.		[2] Mode Array					[3] Address				[4]	[5] Display Data (Page 1)					[5] Display Data (Page 2)					…
P	m	0	1	m	1	32h	22h	FFh	0	0	0	1	1	1	-	20h	1	0	2	-	20h	2	0	

…	[3] Address				[4]	[5] Display Data (Page 1)					[5] Display Data (Page 2)					[5] Display Data (Page 3)				
	0	0	0	2	2	2	-	20h	20h	5	3	-	20h	1	0	5	-	20h	7	0

Explanation

[1] Block number: **01**
[2] Display status: **Specified**
[3] [4] [5] Light module address, page numbers, display data

	Light Module Address	
	0001	0002
1st page display data	"1- 10"	"2- 5"
2nd page display data	"2- 20"	"3- 10"
3rd page display data	None	"5- 70"

Operation

The operation after pressing the **CONFIRM** button or **Fn** key **+ CONFIRM** button

CONFIRM button

Before pressing	**After pressing**	
	Display data	Light module status
If next page exists	Next page	During the operation
Last page	OFF	Normal completion

Fn key

Before pressing the **Fn**	**After pressing**	
	Display data	Light module status
If previous page exists	Previous page	During the operation
1st page	1st page	During the operation

Note

(1) For details on page turning, refer to Section 3. "Page Turning Controllers".
(2) When doing "Repeatable" specify so as not to exceed the maximum command length.

P6 (Page-turning Operation Instruction with Operation Mode)

TW-P | Ethernet | PCI | Repeatable

Function

Give page-turning operation instructions with operation mode to light modules (up to10 pages)

Format

P	6	[1] Block No.	[2]	[3]	[4]	[5]	[6]	[7] Address	[8]	[9] Display Data

No.	Items	Details	Note
[1]	Block No.	Block No.	Specification range: 00 ~ 99
[2]	Display status	d7 d0: LED Color | BUZ | LED | SEG **00**: LED 2, 1, 0 **00**: OFF **01**: LED 2, 0 **01**: ON **10**: LED 1 **10**: Flash **11**: LED 2, 0 **11**: No change	XW LED color is fixed at 01 TW No change if **FFh**
[3]	Display status after pressing the **CONFIRM** button		
[4]	**CONFIRM** button, **Fn** key	d7 d0: 0 | 1 | [a] | [b] | [c] (e) **Fn** key decrement mode **00**: Not use **01**: Use **11**: No change (f) **Fn** key **00**: Use **01**: Not use **11**: No change (c) **CONFIRM** button **00:** Use **01**: Not use **11**: No change	XW [a] **Fn** key decrement mode is fixed at 00 TW No change if **FFh**
[5]	Flashing speed	d7: 0 | 1 | [a] | [b] | 0 | 0 | [c] | 0 [a] Display status after the **CONFIRM** button press [b] Displayed status **0**: Normal [c] Status after the **Fn** Key press **1**: High speed	Flashing speed in case of [2],[3],[6] XW [b] Display status [c] Display status after the **Fn** key is pressed is fixed at 0
[6]	Display status after pressing the **Fn** key	d7 d0: LED Color | BUZ | SEG | LED **00**: LED 2, 1, 0 **00**: OFF **01**: LED 2, 0 **01**: ON **10**: LED 1 **10**: Flash **11**: LED 2, 0 **11**: No change	XW LED colors fixed at 01 BUZ and SEG are fixed at 00 TW No change if **FFh**
[7]	Address	Light module address	Specification range: 0001~7999
[8]	Page	Number of pages	Specification range: 0~9 0: 1 page | 9: 10 pages (Max.)
[9]	Display Data	Display data	Always specify 5 characters per page

Maximum Command Length

Ethernet type	1015 bytes
Board type	1024 bytes
RS-232C type	

Command Examples

(1) Operation mode specification • Page–turning operation instruction

P	6	[1] Block No.	[2]	[3]	[4]	[5]	[6]	[7] Address	[8]	[9] Display Data (Page 1)	[9] Display Data (Page 2)	[9] Display Data (Page 3)
P	6	0 1	77h	FFh	7Fh	40h	FFh	0 0 0 1	2	1 - 20h 1 0	2 - 20h 2 0	3 - 20h 3 0

Explanation

[1] Block number: **01**
[2] Display status: **Specified**
[3] Display status after pressing the CONFIRM button: **Specified**
[4] CONFIRM button, Fn key: **Specified**
[5] Flashing speed: **Specified**
[6] Display status after pressing the Fn key: **Specified**
[7] Light module address: **0001**
[8] Number of pages: **3**
[9] Display data (1st page): **"1- 10",** (2nd page): **"2- 20",** 3rd page **"3- 30"**

(2) Operation mode specification • Page–turning operation instruction

P	6	[1] Block No.	[2]	[3]	[4]	[5]	[6]	[7] Address	[8]	[9] Display Data (Page 1)	[9] Display Data (Page 2)	…
P	6	0 1	77h	FFh	7Fh	40h	FFh	0 0 0 1	1	1 - 20h 1 0	2 - 20h 2 0	

…	[7] Address	[8]	[9] Display Data (Page 1)	[9] Display Data (Page 2)	[9] Display Data (Page 3)
	0 0 0 2	2	2 - 20h 20h 5	3 - 20h 1 0	5 - 20h 7 0

Explanation

[1] Block number: **01**
[2] Display status: **Specified**
[3] Display status after pressing the CONFIRM button: **Specified**
[4] CONFIRM button, Fn key: **Specified**
[5] Flashing speed: **Specified**
[6 Display status after pressing the Fn key: **Specified**
[7][8][9] **Light module address, Number of pages, Display data**

	Light module address	
	0001	0002
Display data, 1st page	"1- 10"	"2- 5"
Display data, 2nd page	"2- 20"	"3- 10"
Display data, 3rd page	None	"5- 70"

Operation

The **CONFIRM** button in operation and operation after pressing the **Fn** key + **CONFIRM** button

CONFIRM button

Before pressing	**After pressing**	
	Display data	Light module status
Next page exists	Next page	Operating
Last page	Off	Normal completion

Fn key + **CONFIRM b**utton

Before pressing	**After pressing**	
	Display data	Light module status
Previous page exists	Previous page	Operating
First page	First page	Operating

Note

(1) Refer to Section 3, “Page-Turning Controller” for more information.

(2) Do not exceed the maximum command length for “Repeatable.”

4.7. Check Completion Command

H

TW | XW | TW-P | PCI | RS-232C

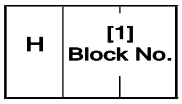
Function

(1) Inquires the status of the block-specified light modules.

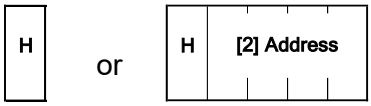
(2) Inquires the status of light modules individually.

Format

(1) Block-specified



(2) Individually specified

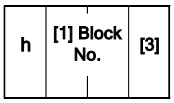
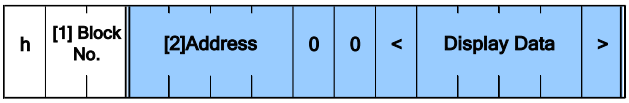


No.	Item	Detail	Note
[1]	Block No.	Block number	Specification range: 00 ~ 99
[2]	Module Address	Inquired light module address	

Operation

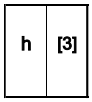
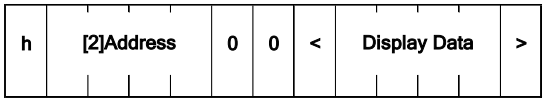
(1) The response of the block completion report of light modules (maximum 128 units) is received with the "**hxx**" command.

Response

	Situation	Detail	Note
①	Where there are no completion light modules	 [3] n: During standby b: During the operation instruction	
②	When there are completion light modules	 [3] Code 00:Normal completion 01:Abnormal light module 10:Shortage	Maximum 128 light modules for "Repeatable"
③	Normal completion when light modules with the +/- key are used after pressing the +/- key, or when the **Fn** key decrement mode is used.		Maximum 128 light modules for "Repeatable" <:3Ch >:3Eh

(2) When the "**h**" command is received, there will be an operation status response.

Response

	Situation	Details	Note
①	When there are no completion light modules	 [3] n: During standby b: During operation instruction	
②	When there are completion light modules	 [3] Code 00: Normal completion 01: Abnormal light modules 10: Shortage	
③	When +/- light modules had normal completion after the +/- operation or when the **Fn** key decrement mode was used		<:3Ch >:3Eh

Command Example

(1) By the Block

H	[1] Block No.	
H	0	1

h	[1] Block No.		[2] Address				[3]		[2] Address				[b]		<	Display Data					>
h	0	1	0	0	0	1	0	0	0	0	0	2	0	0	<	20h	20h	20h	0	9	>

Explanation

At Block No. 01, light module No. 0001 had normal completion, light module No. 0002 had normal completion after the +/- key operation.

[1] Block number: **01**
[2] Light module address: **0001**
[3] Code: **Normal completion**
[4] Light module address: **0002**
[5] Code: **Normal completion**

Note

Unless **"Use the +/- key"** is specified in the **@** command during page-turning when quantity is revised using the light module's **+/-** key or the **Fn** key decrement mode, the application cannot capture the revised result**<Display data>**. The result of the revision **<Display Data>** is shown below

<	Page	-	Display Data	>
3Ch	0-9 0=1Page 9=10Page	2Dh	5Byte	3Eh

For more information, see "**3. Page-turning Controller**".

The Polling Cycle to the Controller to get the Completion Confirmation with the "H" Command

Because the PC speed has dramatically increased, it is necessary to be aware of the polling cycle when using the board type or RS-232C type controller. The optimum speed is between 80~120ms. When the polling cycle is less than 80ms with the **"H"** command, the controller's processing ability is actually reduced and impacts the overall speed. If the cycle is longer than 120ms, the controller is not affected but will impact the overall speed. Also, when a system has many blocks, control of application may be easier with the block configuration (e.g., Block No.). However, if the block specification **"H"** command is used for the completion confirmation and if it takes 100 ms per block, the total time for all blocks is 100ms times the number of blocks. This will greatly affect the overall speed.

In this case, use one of the following but AIOI Systems recommends (2):

(1) Use (Not Use Block Number) of the **"H"** command. Light module completion is obtained one at a time.
(2) Do not use multiple block numbers with operation instruction. Specify all with **"00"**.
Use **"H00"** for the polling completion confirmation.

It is not necessary to consider the polling cycle to find out if barcode data exists or not.
However, monitor in shortest intervals as possible.

4.8. Termination Instruction Command

D (Terminate Operation of Specified Modules)

TW | TW-P | XW | Ethernet | PCI | RS-232C | Wildcard

Function

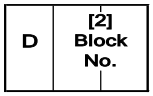
(1) Terminates the module's operation instruction specified by the application.
(2) Terminates the light module operation instruction within the Block No. specified by the application and turns OFF the signal light(or batch display.

Format

(1) Specified by the module. Repeatable



(2) Specified by the block



No.	Item	Detail	Note
[1]	Address	Module addresses to be terminated	
[2]	Block No.	Block No. to be terminated	Even when a wildcard character is used, it will not turn OFF the light modules with the display status "ON/Flashing" after pressing the **CONFIRM** button.

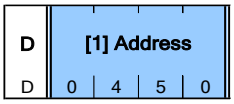
Maximum Command Length

For (1)

Ethernet type	128 addresses
Board type	
RS-232C type	

Command Example

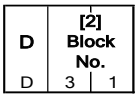
(1) Specify by the module



Explanation

[1] Address of the module to be terminated: **450**

(2) Specify by the block



Explanation

[2] Block number to be terminated: **31**

Operation

Below is the operation of various modules when this "**D**" command is sent.

Modules	Operation
Light module, Signal light, Batch display	Turn OFF, Operation is terminated
One point I/O	OFF is outputted
Barcode reader interface	No data transmission
RS-232C interface	No data transmission

Note

(1) When specified by the block, the light modules won't turn OFF with the display status "ON/Flashing" after pressing the **CONFIRM** button.

(2) For "Repeatable," do not exceed the maximum command length.

4.9. Batch Display Command

Batch Display Command List

Command	Details	Applicable		Page No.
		TW	XW	
Am1	Specify batch display operation mode	○		78
X	Send display code to the batch display	○	○	79
Q2	Send display code to the batch display	○		80

○: Applicable

Am1 (Batch Display Operation Mode Specification)

TW | TW-P | Ethernet | PCI | RS-232C | Repeatable | Wildcard

Function

Specifies the operation mode of the batch display.

Format

A	m	1	[1] Address	[2]	11h	[3]

No.	Item	Detail
[1]	Address	Batch Display Address
[2]	Scroll mode	d7 d0 [a] 0 0 0 1 **[a] Scroll Mode** **0001:** Scroll function **0010:** Scroll function (pause) **0011:** No scroll function
[3]	Display mode scroll speed setup	d7 d0 [b] [c] **[b] Display mode** **0001**:OFF **0010**: ON **0011**: Flash **0100**: High speed flash **1111**: No change **[c] Scroll speed** **0001**: Normal **0010**: High speed **0100**: Fastest **1111**: No change:

Maximum Command Length

Ethernet type	64 addresses
Board type	
RS-232C type	

Command Example

A	m	1	[1] Address				[2]	[3]	[4]
A	m	1	8	1	?	?	21h	11h	21h

Explanation

[1] Batch display address: **8100 ~ 8199**

[2] Scroll function: **Scroll with a pause**

[3] Display mode, scrolling speed setup, display mode: **ON**; scroll speed: **Normal**

Note

(1) Send this command before sending the display code to the batch display.

(2) For “Repeatable,” do not exceed the maximum command length.

X (Display Code Transmission to the Batch Display)

TW | TW-P | XW | Ethernet | PCI | RS-232C | Wildcard

Function

Sends the display code to the batch display. (Standard command, 12 fixed characters)

Format

Sending the display code

X	[1] Address	[2] Display Data

No.	Item	Detail	Note
[1]	Address	Batch display address	
[2]	Display Data	Display code	Display code is left justified. No zero suppression of the display code. Use the ASCII code for the display code.

Command Example

Sending the display code

X	[1] Address				[2] Display Data										
X	8	1	0	1	L	I	G	H	T	20h	M	O	D	U	L	E

Explanation

[1] Batch display address: **8101**

[2] The display code: "**LIGHT△MODULE**" ("△" is a space)

Q2 (Display Code Transmission to the Batch Display)

TW | TW-P | Ethernet | PCI | RS-232C | Wildcard

Function

The display code up to 20 characters are sent to the batch display.

Format

Sending the display code

Q	2	[1] Address	[2] Data Length	[3] Display Data

No.	Item	Detail	Note
[1]	Address	Batch display address	
[2]	Data Length	Data length	Specification range: 04h ~ 14h (Hex)
[3]	Display Data	Display code	Number of bytes in the data length Specify the display code in ASCII code

Command Example

Sending the display code

Q	2	[1]Address				[2] Data Length	[3] Display Data															
Q	2	8	1	0	1	14h	O	R	A	N	G	E	20h	J	U	I	C	E	20h	2	0	0	m	l	*	6

Explanation

[1] Batch display address: **8101**

[2] Data length: **14h**

[3] Display code: "**ORANGEΔJUICEΔ200ml*6**" ("Δ" is a space)

Supplement

To display 20 characters on the 10- digit batch display:

- "**Am1**" command scroll mode: **Scroll function**
- "**Q2**" command: **20 characters**

When specified as above, all 20 characters can be seen right to left.

4.10. One-Point I/O Command

One-point I/O Command List

Command	Details	Applicable		Page No.
		TW	XW	
Am1	Specify one-point I/O operation mode	○		82
M	Specify one-point I/O operation mode	○		83
T	Turn ON/OFF one-point I/O	○		84

○: Applicable

Am1 (One-point I/O Operation Mode Specification)

TW | TW-P | Ethernet | PCI | RS-232C | Repeatable | Wildcard

Function

Specifies the one-point I/O operation mode.

Format

A	m	1	[1] Address	32h	22h	11h

No.	Item	Detail	Note
[1]	Address	One-point I/O address	Specification range: 1~7999

Maximum Command Length

Ethernet type	64 addresses
Board type	
RS-232Ctype	

Command Example

A	m	1	[1] Address	32h	22h	11h
A	m	1	0 0 0 3	32h	22h	11h

Explanation

[1] Operation mode is specified on the one-point I/O address: **0003**

Note

(1) Always send the initialization setup command "**Z**" before sending the "**Am1**" command.

(2) For "Repeatable", do not exceed the maximum command length.

M (One-point I/O Operation Mode Specification)

TW | TW-P | Ethernet | PCI | RS-232C | Wildcard

Function

Specifies the operation mode on the one-point I/O.

Format

M	[1] Address	04h	30h	30h	40h	30h

No.	Item	Detail	Note
[1]	Address	One-point I/O address	Specification range: 1~7999

Command Example

M	[1] Address				04h	30h	30h	40h	30h
M	1	0	0	1	04h	30h	30h	40h	30h

Explanation

[1] The operation mode is specified on the one-point I/O address: **1001**

Note

Always send after the initialization setup command **"Z"**.

T (Turn ON/OFF of One-point I/O)

TW | TW-P | Ethernet | PCI | RS-232C | Wildcard

Function

Turn ON or OFF the one-point I/O.

Format



No.	Item	Detail	Note
[1]	Address	One-point I/O address	Specification range: 1~7999
[2]	Switch (SW)	0: OFF output 1: ON output	

Command Example



Explanation
[1] One-point I/O address: **1001**
[2] Turn the switch (SW): **OFF**

Note

The one-point I/O has one-point output and one-point input. The method of output is described here. For information on the input, contact AIOI Systems Co. Ltd.

4.11. (Intelligent) Signal Light (Tower) Command

(Intelligent) Signal Light (Tower) Command List

Command	Detail	Acceptable		Page No.
		TW	XW	
Am1	Specify signal light operation mode	○		86
M	Specify signal light operation mode	○		87
T	Turn ON/OFF signal light	○	○	88

○: Applicable

Am1 (Signal Light Operation Mode Specification)

TW | TW-P | Ethernet | PCI | RS-232C | Repeatable | Wildcard

Function

Specifies the operation mode on (intelligent) signal light (tower).

Format

A	m	1	[1] Address	31h	[2]	[3]

No.	Item	Detail	Note
[1]	Address	Signal light address	Specification range: 8001~8099
[2]	Light	11h : OFF 12h : ON 13h : Flashing 14h : High speed flashing	
[3]	Buzzer (For intelligent light tower only)	11h : OFF 12h : ON 13h : Flashing 14h : High speed flashing	

Maximum Command Length

Ethernet type	64 addresses
Board type	
RS-232C type	

Command Example

A	m	1	[1] Address				31h	[2]	[3]
A	m	1	8	0	0	1	31h	13h	11h

Explanation

[1] Signal light address: **8001**
[2] Light: **Flash**
[3] Buzzer: **OFF**
Operation mode specified.

Note

For “Repeatable”, do not exceed the maximum command length.

M (Signal Light Operation Mode Specification)

TW | TW-P | Ethernet | PCI | RS-232C | Wildcard

Function

Specifies the operation mode of the (intelligent) signal light tower.

Format

M	[1] Address	[2]	0	0	[3]	0

No.	Item	Detail	Note
[1]	Address	Address of the signal light	Specification range: 8001~8099
[2]	Display status	d7 d0 0 0 BUZ Light 0 1 Only for the intelligent 00 : OFF 00 : OFF 01 : ON 01 : ON 10 : Flash 10 : Flash	For the Ethernet, PCI Specify so that the result is not **02h** or **03h.** For the RS232C Set up so that the result is not **0Dh**.
[3]	Flashing speed	d7 d0 0 1 0 0 0 0 0 0: Standard 1: High speed	

Command Example

M	[1] Address				[2]	0	0	[3]	0
M	8	0	?	?	09h	30h	30h	60h	30h

Explanation

[1] Signal light address: **8001~8099**
[2] Display status: **Flashing**
[3] Flashing speed: **High speed**

T (Turn the Signal Light ON/OFF)

TW | TW-P | XW | Ethernet | PCI | RS-232C | Wildcard

Function

Turns the signal light ON or OFF.

Format

T	[1] Address	[2]

No.	Item	Detail	Note
[1]	Address	Signal light address	Specification range: 8001~8099
[2]	ON/OFF specification	0: OFF 1: ON	

Command Example

T	[1] Address				[2]
T	8	0	0	1	1

Explanation

[1] Signal light address; **8001**
[2] ON/OFF specification: **ON** (turn light ON)

4.12. Barcode Reader Interface, RS-232C Interface Command

Barcode Reader Interface Command List

Command	Details	Applicable		Page No.
		TW	XW	
G-o , c	Instruction to the barcode reader interface/RS-232C interface	○	○	90
G-t	Set up the reception priority of barcode data and light module completion	○	○	91
G-i	Data transmission to the device connected to the interface	○	○	92
Ar	Check RS-232C interface status	○		93

○: Applicable

G-o,c (Instruction to the Barcode Interface/RS-232C Interface)

TW | TW-P | XW | Ethernet | PCI | RS-232C | Wildcard

Function

Instruction to Open /Close various interfaces.

Format

G	[1]Address	[2]

No.	Item	Detail			Supplement
[1]	Address	Interface address			
[2]	Instruction details	o	Barcode reader interface	Open (start monitoring)	Supply electrical power to the scanner
			RS-232C interface	Start receiving	
		c	Barcode reader interface	Close (stop monitoring)	Stop the electrical power supply to the scanner
			RS-232C interface	Stop receiving	

Command Example

G	[1]Address	[2]
G	9 1 0 1	o

Explanation

Barcode reader interface No. **9101** is **open** (An instruction to read a barcode with a scanner)

Operation

(1) Barcode reader interface

The scanner connected to the barcode reader interface starts (OPEN) or stops (CLOSE) data capture. When set to OPEN, power is supplied to the scanner.

(2) RS-232C interface

Begins or stops the data reception from the barcode printer connected to the interface.

Note

(1) The default is CLOSE.

(2) When instructed to OPEN, each interface will clear the reception buffer from the connected devices, such as a scanner.

(3) If a barcode is not going to be scanned, giving the “CLOSE” instruction is an effective method from the system point of view. However, from the timing of the instruction for scanning and CLOSE, the barcode data may show up after “CLOSE”. With the application, delete the barcode data.

G-t (Set up the Barcode Data Reception Priority)

TW | TW-P | XW | Ethernet | PCI | RS-232C

Function

The controller begins monitoring when the "**G-o**" command (Open) is sent to the barcode reader interface and RS232C interface. The "**G-t**" command sets the monitoring priority with the light modules that received instructions,

Format

G	9	1	?	?	t	[1]

No.	Item	Detail	Supplement
[1]	Monitoring interval of the interface	After monitoring the setup values of light modules, monitors all open interfaces.	01-99 BCD of ASCII Default= 05

Command Example

G	9	1	?	?	t	[1]	
G	9	1	?	?	t	1	0

Explanation

To do the priority capture of the light module completion status (by pressing the **CONFIRM** button), set up so that first the **10** light modules are monitored, then the interface is monitored.

Operation

Use when the setup is other than the default.

Note

Normally, it is not necessary to set up.

G-i (Data Transmission to the Device Connected to the Interface)

TW | TW-P | XW | Ethernet | PCI | RS-232C | Wildcard

Function

Transmit data to the devices connected to the interface.

Format

G	[1]Address	i	[2] Data Length	[3]Data

No.	Item	Detail		Supplement
[1]	Address	The interface address of the connected device to which data is sent.		
[2]	Data Length	Barcode reader interface	001E	When sending, specify the Data length in ASCII binary
		RS-232C interface	00-F8	
[3]	Data	Set up the data on the byte length specified in [2].		

Command Example

G	[1]Address				i	[2] Data Length		[3]Data										
G	9	1	0	1	i	0	b	1bh	2	0	0	5	/	0	1	/	1	0

Explanation

[1] To the barcode printer connected to the address **"9101"** of the RS-232C interface, send
[2] **0b** (11 bytes) and
[3] **1bh** (in Esc sequence) print data **"2005/01/10"**

Operation

(1) In the case of the barcode reader interface, this command is used when sending the barcode scanned data.

(2) In the case of the RS−232C interface, this command is used when sending the print data to the connected barcode printer, etc.

Note

(1) When [2]Data Length is **"00"**, do not set the data for "[3]Data".

(2) This command can be sent during both "Open" and "Close".

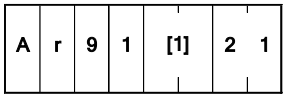
Ar (RS-232C Interface Status Check)

TW | TW-P | Ethernet | PCI | RS-232C

Function

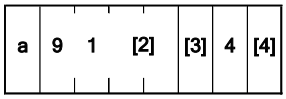
The "**Ar**" command checks the status of the RS-232C interface.

Format



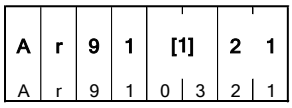
No.	Item	Detail	Supplement
[1]	Address	Specify the last 2 digits of the RS-232C interface	ASCII's 01-99 BCD

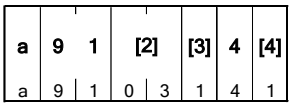
Response



No.	Item	Detail			Supplement
[2]	Address	Last 2 digits of the specified RS-232C interface			
[3]	RS-232C Interface status 1	1 (31h)	OK (normal)		When "NG," the next 2 bytes (4,[4]) do not exist
		? (3Fh)	NG (cannot control with the application)		
[4]	RS-232C interface status 2	D7	Undefined		
		D6	Undefined		
		D5	Undefined		
		D4	Undefined		
		D3	RS-232C DTR	0	Ready
				1	Busy
		D2	RS-232C interface's buffer	0	OK
				1	Near Full
		D1	Barcode printer is Ready (RS-232C RS)	0	Ready
				1	Busy
		D0	Barcode printer's power	0	OFF
				1	ON

Command Example





Explanation

[1] Print data is sent to **9103** of the RS-232C interface, and the status is inquired.
[2] RS-232C interface No. **9103**
[3] RS-232C interface status 1: **OK**
[4] The "G-I" command can be sent – Barcode printer's power: **ON**, Barcode printer: **Ready**, RS-232C interface buffer: **OK**,
RS-232C DTR: **Ready**

Operation

(1) Returns the status of: RS−232C interface, RS−232C, and internal buffer.

(2) Check the status before sending data to the connected devices, such as a barcode printer.

Note

When the RS-232C interface buffer is **Near Full** ("**D2**" of **[4]**), there may be a connection problem with the interface. The "**G-i**" command cannot be sent, and the internal buffer has become almost full.
The data may be damaged if sent under this condition.

4.13. Report from the Controller

Report Related Commands from the Controller

Command	Detail	Applicable		Page No.
		TW	XW	
t	Response data of each device	○		96
O	Response (ACK) from the application	○		100
N	Negative response (NACK) from the application	○		101

○: Applicable

t (Response Data of the Device)

TW | TW-P | Ethernet | PCI | RS-232C

Function

The controller sends the "**t**" response when the **CONFIRM** button of the light module is pressed, or when there is an operation change, such scanning a barcode and reading the barcode data. The "**t**" response is sent when:

- An operation has been completed (light module completion)
- A scanner connected to the open barcode reader interface has read the barcode data
- Data is received from a module connected to the open RS-232C interface
- Data is inputted by the 10-key pad of the open RS-232C interface

How to Identify a Module

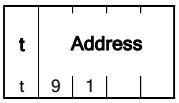
The first 2 digits differentiate the modules.

(1) If the first 2 digits are not 91, it is light module. Repeatable , Ethernet



(2) If the first 2 digits are 91, it is a barcode reader interface or RS-232C interface.

Ethernet , PCI , RS-232C



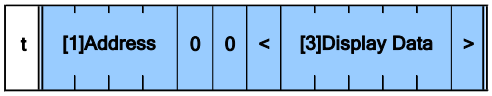
Format

(1) Operation completion (Light module completion) Repeatable , Ethernet

- The quantity is not revised



- Quantity revised using the light module **+/-** key or the **Fn** key decrement mode

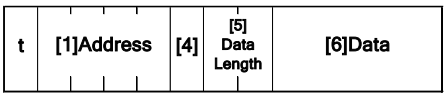


Maximum Command Length

1015 bytes

(2) Barcode data (barcode reader interface, RS-232C interface)

＊Not **Repeatable**



No.	Item	Detail		Supplement
[1]	Address	Address of the module that received a response		
[2]	Status	00	Normal completion	
		01	Abnormal	
		10	Shortage	
[3]	Display Data	Revised quantity		
[4]	Status2	1	Normal	0 (there is no [5] or [6] when abnormal)
		0	Abnormal	
[5]	Data Length	01-30		Set the [6]Data length as BCD in ASCII
[6]	Data	Data from the scanned barcode or data from modules connected to the interface		**01h-1Fh** is delimiter of the interface. The delimiter is not included in the data.

Command Example

(1) Operation completed (light module completion)

・ Quantity is not revised.

t	[1]Address				[2] Status		[1]Address				[2] Status	
t	1	1	0	1	0	0	2	0	0	1	1	0

Explanation

[1] Light module: **1101**

[2] **Normal completion**

[1] Light module: **2001**

[2] **Shortage**

・ Quantity revision after using the **+/-** key of the light module with the **+/-** key or the **Fn** key decrement mode

t	[1]Address				0	0	<	[3] Display Data					>
t	0	1	0	1	0	0	3Ch	20h	20h	20h	1	5	3Eh

Explanation

[1] Light module: **0101**

[3 Normal completion after the quantity was revised to **15**

(2) Barcode data (barcode reader interface, RS-232C interface)

t	[1]Address				[4]	[5] Data Length		[6] Data										
t	9	1	0	1	1	1	3	4	9	0	3	0	1	5	1	1	0	8	1	7

Explanation

[1] With the scanner connected to the barcode reader interface **9101**

[5] **13** digits of

[6] JAN code was scanned

Operation

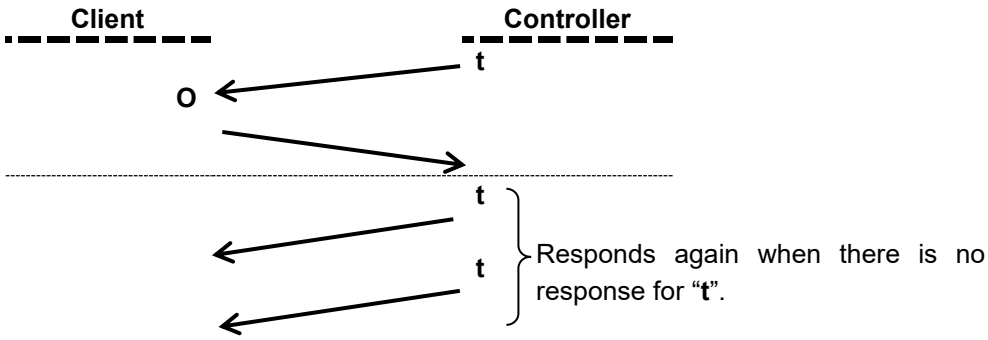
Always send the "**O**" command.

(1) This "**t**" command is sent from the controller when the light module operation is finished (**CONFIRM** button, **Fn** key + **CONFIRM** button) or when there is malfunction.

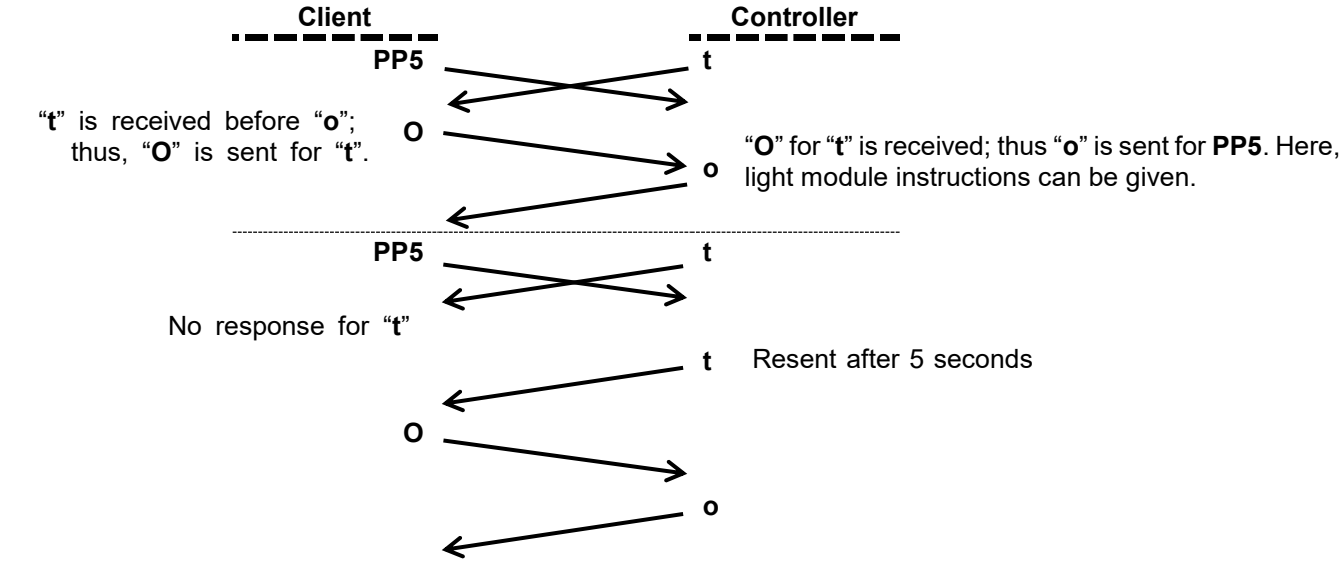
(2) This command is sent from the controller after a barcode is scanned, or the RS-232C interface receives the data from the connected device, or there is an RS-232C interface's 10-key input.

Sequence

[Basic Flow]



[When Client and Controller both send at the same time]



Note

(1) Network malfunction: cable, connector, devices
(2) Session is severed
(3) Took more than 5 seconds for some kind of processing
(4) The correct "**O**" was not sent
(5) Did not follow the above sequence

O

TW | TW-P | Ethernet | PCI | RS-232C

Function

When the PC receives the report command "**t**" from the controller, the PC returns "**O**" as a successfully received Response.

Format

o

Note

(1) Immediately send "**O**" when there is a report from the controller.
(2) The controller will not send the next command until it receives a response from the application.

N (Negative Response Sent from the Application)

TW | TW-P | PCI | RS-232C

Function

The "**N**" command is sent when the controller's transmitted data is not received correctly.

Format

N

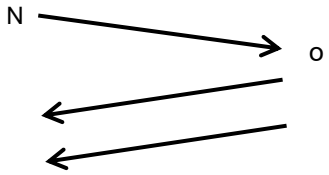
Operation

Sends the data that was previously sent to the application.
The following can be considered as to why the application sent a negative response.

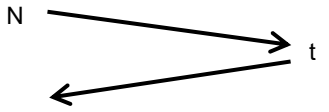
(1) Not enclosed with "**STX**" or "**ETX**" (in the case of board type).
(2) The checksum is incorrect. (in the case of RS-232C)
(3) There is a mistake in the data length (the data length and number of data do not match).

When the "**N**" command is sent, however, there are 2 sequences and 3 possible meanings.

(1) If this response is a response that has not been processed, the earlier one is considered as a communication problem and processes the current one.
(2) If this response has already been processed, it concludes that somehow it received garbage and resumes.



(3) If it is a "**t**" command, it thinks that the previous "**t**" command was a communication error and processes the current "**t**" command.



In another words, send "**N**" if you want the controller to respond again.

Note

To determine whether a response was received correctly, wait for 3 seconds.

4.14. Digital I/O Control Command

Digital I/O Control Command List

Command	Detail	Applicable		Page No.
		TW	XW	
J	Digital I/O output, 8-bit type	○	○	103
J	Digital I/O output, 16-bit type	○	○	104
E	Digital I/O input	○	○	105
Ai	Read digital I/O output data	○		106

○: Applicable

J (Digital I/O Output) 8-bit type

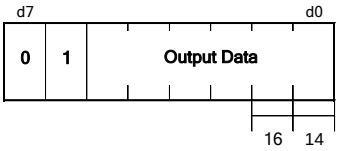
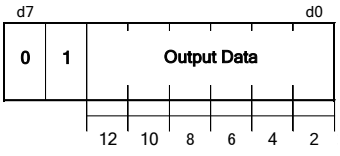
TW | TW-P | XW | Ethernet | PCI | RS-232C | Wildcard

Function

Specifies the data output to the digital I/O.

Format

J	[1] Address	30h	30h	30h	30h	[2]	[3]

No.	Item	Detail	Note
[1]	Address	Digital I/O address	
[2]	Output data		Output logic is according to the Client's output.
[3]	Output data		Output logic is according to the Client's output.

Command Example

J	[1] Address				30h	30h	30h	30h	[2]	[3]
J	7	1	0	0	30h	30h	30h	30h	40h	54h

Explanation

[1] Digital I/O address: **7100**

[2] Terminal plate number: **Low** for **16, 14**

[3] Terminal plate number: **High** for **6**, **10**, **Low** for **all others**

J (Digital I/O Output) 16-bit type

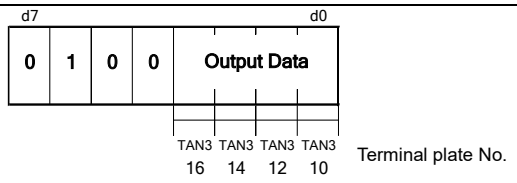
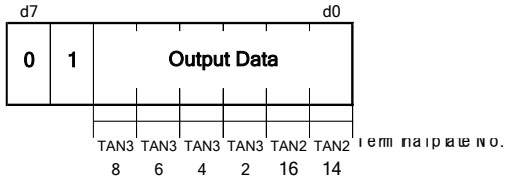
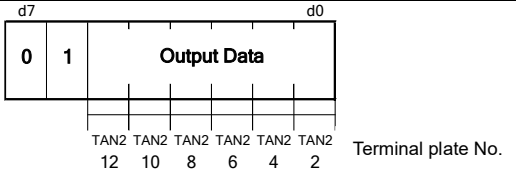
TW | TW-P | XW | Ethernet | PCI | RS-232C | Wildcard

Function

Specifies the data output to the digital I/O.

Format

J	[1] Address	30h	30h	30h	[2]	[3]	[4]

No.	Item	Detail	Note
[1]	Address	Digital I/O address	
[2]	Output Data		The output logic is the same as the output of the client. The LED of the output terminal monitor will turn **ON** when "**0**" is output.
[3]	Output Data		The output logic is the same as the output of the client. The LED of the output terminal monitor will turn **ON** when "**0**" is output.
[4]	Output Data		The output logic is the same as the output of the client. The LED of the output terminal monitor will turn **ON** when "**0**" is output.

Command Example

J	[1] Address				30h	30h	30h	[2]	[3]	[4]
J	7	1	0	0	30h	30h	30h	30h	40h	54h

Explanation

[1] Digital I/O address: **7100**

[2] TAN3: Terminal plate numbers **16**, **14**, **12**, **10** are **Low**

[3] TAN3: Terminal plate numbers **8**, **6**, **4** and **2** are **Low**

[4] TAN2: Terminal plate numbers **6** and **10** are **High**, **other terminals** are **Low.**

E (Digital I/O Input)

TW | TW-P | XW | Ethernet | PCI | RS-232C

Function

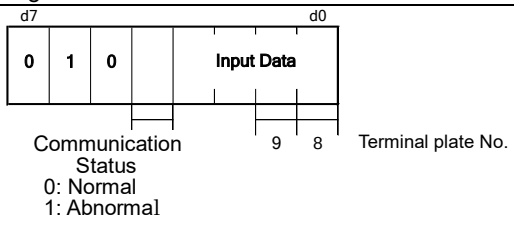
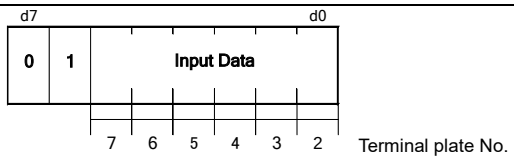
Capture the input condition from the digital I/O (8-bit)

Format

E	[1] Address

Response

e	[1] Address	[2]	[3]	[4]

No.	Item	Detail
[1]	Address	Digital I/O address
[2]	Input data	
[3]	Input data	
[4]	Undefined	

Command Example

E	[1] Address			
E	1	5	0	0

Explanation

[1] Digital I/O address: **1500**

e	[1] Address				[2]	[3]	[4]			
e	1	5	0	0	40h	47h	40h	40h	40h	40h

Explanation

[1] Digital I/O address: **1500**

[2] Communication status: **Normal**, Terminal plate numbers **9**, **8** are **Low**

[3] Terminal plate numbers: **2**, **3**, **4** are **High, all others** are **Low**

[4] Undefined: Can be **High** or **Low.**

Note

The digital I/O data output and input will not be stored in the controller.

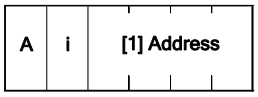
Ai (Reading Digital I/O data Output)

TW | TW-P | Ethernet | PCI | RS-232C

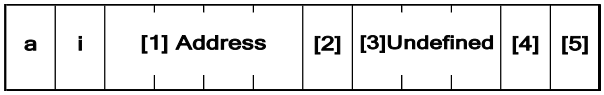
Function

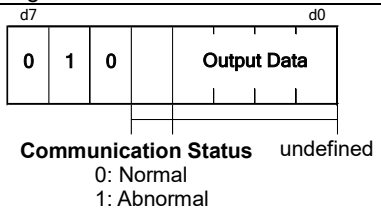
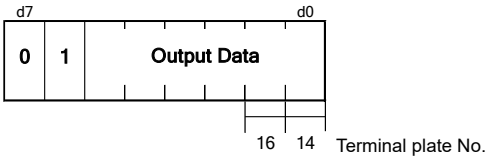
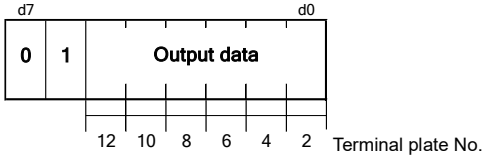
Read the digital I/O data output.

Format



Response



No.	Item	Detail
[1]	Address	Digital I/O Address
[2]	Output data	
[3]	Undefined	Undefined
[4]	Output data	
[5]	Output data	

Operation

When the "**Ai**" command is sent, the digital I/O returns the last data sent by the output buffer, and a comparison is made between this last data and the data sent by the I/O output command "**J**" to determine whether the output data was sent correctly to the digital I/O.

Command Example

A	i	[1] Address			
A	i	1	2	0	0

a	i	[1] Address				[2]	[3]Undefined			[4]	[5]
a	i	1	2	0	0	40h				40h	48h

Explanation

[1] Digital I/O address: **1200**

[2] Communication: **Normal**

[3] Undefined: Can be **any value**

[4] Terminal plate No.: **14** and **16** are **Low**

[5] Terminal plate No. **8** is **High**, **others** are **Low**

4.15. Light Module Display Command

Light Module Display Command List

Command	Detail	Applicable		Page No.
		TW	XW	
PP2	Light module display	○		109
P2	Light module display	○	○	111
FP2	Light module display, TW2060 type	○		112

○: Applicable

PP2 (Light Module Display)

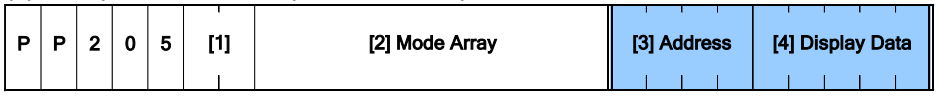
TW | TW-P | Ethernet | PCI | RS-232C | Repeatable | Wildcard | MA

Function

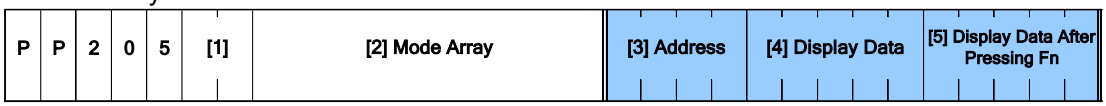
Flashes the light modules.

Format

(1) Operation mode specification・Operation instruction



(2) Operation mode specification・Operation instruction・Display data specification after pressing the **Fn** key



No.	Item	Detail	Note
[1]	Display data specification after pressing the Fn key	00: Not specified 05: Specified	00: [5] is omitted 05: 5-character display data after pressing the Fn key is specified at [5]
[2]	Mode Array	Refers to “Mode Array” Variable length, Can be omitted	When omitting the Mode Array, continue from “Address”.
[3]	Address	Module address	
[4]	Display Data	Display data	Always specify 5 characters.
[5]	Display Data After Pressing **Fn**	Display data after pressing the **Fn** key	Omit when [1] is 00. Always specify 5 characters when [1] is 05.

Maximum Command Length

1015 bytes

Command Example

(1) Operation mode specification・Operation instruction

P	P	2	0	5	[1]		[2]Mode Array				[3] Address				[4] Display Data				
P	P	2	0	5	0	0	m	2	32h	22h	21h	1	0	0	1	20h	20h	20h	20h	5

Explanation

[1] Display data specification after pressing the Fn key: **Not specified**

[2] Display status after pressing the CONFIRM button: **Specified**

[3] Module address: **1001**

[4] Display data: " **5**" (space, space, space, space, 5)

(2) Operation mode specification・Operation instruction・Display data specification after pressing the **Fn** key

P	P	2	0	5	[1]		[2] Mode Array									[3] Address				[4] Display Data					[5]Display Data After Pressing Fn				
P	P	2	0	5	0	5	m	1	34h	44h	FFh	m	3	3Fh	FFh	3Fh	1	0	0	1	20h	20h	20h	20h	8	6	6	6	6	6

Explanation

[1] Display data specification after pressing the Fn key: **Specified**

[2] Display status: **Specified**, Display status after pressing the Fn key: **Specified**

[3] Module address: **1001**

[4] Display data: " **8**"

[5] Display data after pressing the Fn key: **66666**

Note

(1) When sending an operation instruction command to the light modules that are displaying, first send the termination command "**D**".

(2) When an address of the module is specified with "**????**", the light module may not display properly. When sending to all the light modules, specify "**0??? ~ 7???**".

(3) For "**Repeatable**", do not exceed the maximum command length.

Supplement

The "**PP2**" command is used for modules that do not require termination (normal completion/shortage). Because when the **CONFIRM** button is pressed, the display turns OFF (default), we recommend specifying the **CONFIRM** button / **Fn** key to "**Not use**" (**Am4**, **M**).

P2 (Light Module Display)

TW | TW-P | XW | Ethernet | PCI | RS-232C | Repeatable | Wildcard

Function

Lights up the light module.

Format

P	2	[1] Address	[2] Display Data

No.	Item	Detail
[1]	Address	Module address
[2]	Display Data	Display data

Maximum Command Length

Ethernet type	1015 bytes
Board type	128 addresses
RS-232C type	

Operation

The operation mode will operate (display) with the mode already specified.

Command Example

P	2	[1] Address				[2] Display data					[3] Address				[4] Display Data				
P	2	0	1	0	0	20h	20h	20h	5	0	0	2	0	0	20h	20h	20h	3	0

Explanation

[1] Module address: **0100**

[2] Display data: " **50**" (space, space, space, 50)

[3] Module address: **0200**

[4] Display data: " **30**" (space, space, space, 30)

Note

(1) If an operation instruction command is going to be sent to light modules that are displaying, first send the termination command "**D**".

(2) When an address of the module is specified with "**????**", the light module may not display properly. When sending to all the light modules, specify "**0???? ~ 7???**".

(3) For **"Repeatable"**, do not exceed the maximum command length.

Supplement

The "**P2**" command is used for modules that do not require completion (normal completion /shortage). Because when the **CONFIRM** button is pressed, the display turns OFF (default), we recommend specifying the **CONFIRM** button / **Fn** key to "**Not use**" (**Am4**, **M**).

FP2 (Light Module Display for TW2060)

TW | TW-P | Ethernet | PCI | RS-232C | Repeatable | Wildcard | MA

Function

Display data length is specified to light up the light modules.

Format

(1) Operation mode specification・Display instruction

F	P	2	0	6	0	0	[2] Mode Array	[3] Address	[4] Display Data

(2) Operation mode specification・Display instruction・Display data specification after pressing the **Fn** key

F	P	2	0	6	[1]	[2] Mode Array	[3] Address	[4] Display Data	[5] Display Data After Pressing Fn

No.	Item	Detail	Note
[1]	Display data specification after pressing the **Fn** key	Data length of the display data after pressing the **Fn** key	
[2]	Mode Array	Refer to “Mode Array” Variable length Can be omitted.	
[3]	Address	Light module address	Specification range: 0001~7999
[4]	Display Data	Display data	Always specify 6 characters
[5]	Display Data After Pressing **Fn**	Display after pressing the **Fn** key	Specify the data length [1] of the character length.

Maximum Command Length

1015 bytes

Command Example

(1) Operation mode specification・Display instruction

F	P	2	0	6	0	0	[2] Mode Array					[3] Address				[4] Display Data					
F	P	2	0	6	0	0	m	1	32h	22h	21h	1	0	0	1	1	2	3	4	5	6

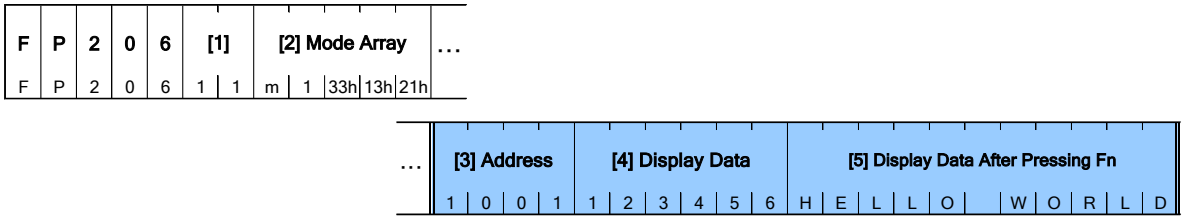
Explanation

[2] Display status: **Specified**

[3] Light module address: **1001**

[4] Display data: "**123456**"

(2) Operation mode specification・Display instruction・Display data specification after pressing the **Fn** key



Explanation

[1] Display data length after pressing the Fn key: **11**

[2] Mode array: Mode specified with “**m1**”

[3] Light module address: **1001**

[4] Display data: "**123456**"

[5] Display data after pressing the Fn key: "**HELLO WORLD**"

Note

For “**Repeatable**”, do not exceed the maximum command length.

4.16. Maintenance Command

Maintenance Command List

command	Detail	Applicable		Page No.
		TW	XW	
A	Specifies maintenance mode	○	○	115
Az	Inquires module status	○	○	116
Ac	Set up module address	○		118

○: Applicable

A

TW | TW-P | XW | Ethernet | PCI | RS-232C

Function

Specifies maintenance mode

Format

A

Operation

When the command "**A**" is sent, the modules will operate as follows:

Modules	Operation
Light module	(1) "**88888**" will be displayed in the segment. (2) Displays a 4-digit self-address after pressing the **CONFIRM** button. (3) Each time the **CONFIRM** button is pressed, the self-address is displayed digit by digit. The punctuation is indicated by '**-**'.
Batch display	Displays its self-address.
Barcode reader interface	Displays the last 2 digits of its self-address.
RS-232C interface	

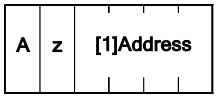
Az (Module Status Inquiry)

TW | TW-P | XW | Ethernet | PCI | RS-232C

Function

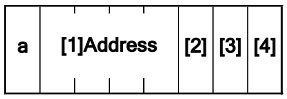
Inquiries the status of modules.

Format



Response

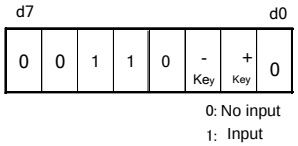
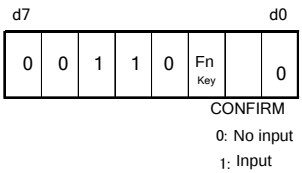
(1) Light module response



(2) Barcode reader interface, RS-232C interface, duct type batch display, light module response from certain series or types*** (1)**



***(1)** SW・AW・DW・XW series light modules, TW2060 type, and "poka yoke" (error detection) light modules such as TW2916

No.	Item	Detail		Note
[1]	Address	Address of the module that received the status inquiry		
[2]	Module status	30h	Immediately after the power supply	Before sending the "**Z**" command
		31h	During standby	After sending the "**Z**" command or after the operation completion
		32h	During the operation instruction	No buttons have been pressed.
		33h	During the operation instruction	The **Fn** key is being pressed
		34h	Operation completion	Normal completion
		35h	Shortage	Canceled (**Fn** key **+ CONFIRM** button)
		36h	During maintenance	The "**A**" command sent
		3Fh	Module malfunction	Disconnected or damaged
[3]	Switch Input status 1			"**Input**" when the "**Az**" command is sent while pressing the "**+**" key or "**-**" key.
[4]	Switch Input status 2			Input status when the "**Az**" command is sent while the **Fn** key and **CONFIRM** button are pressed.

Command Example

A	z	[1]Address			
A	z	1	0	1	5

a	[1]Address				[2]	[3]	[4]
a	1	0	1	5	1	0	0

Explanation
Inquired the status of light module No. **1015**, none of the switches was pressed, and was **normal**.

A	z	[1]Address			
A	z	9	1	0	1

a	[1]Address				[2]
a	9	1	0	1	2

Explanation
When the status of the barcode reader interface No. **9101** was inquired during the open instruction (**G-o** command), it was **during the operation**.

Operation

(1) Can get the status of the specified address at that moment.

(2) The "**H**" command responds to the status of the light module controlled by the controller, and the "**Az**" command returns the status of the specified light module at that moment.

Note

The "**Az**" command will return the status of the module; however, use this command for maintenance.

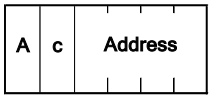
Ac (Set Up the Module Address)

TW | TW-P | XW | Ethernet | PCI | RS-232C

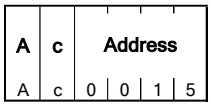
Function

Set up the specified module address.

Format



Command Example



Explanation
Set up the light module address **15**.

Operation

Set up the address by sending the "**Ac**" command to the module while the "switch" is pressed.

Modules	Switch	When the Setup is Completed
Light module	CONFIRM	Peep sound and the LED turns ON instantly
Batch display (duct type)	Address setup switch	No change
Barcode reader interface	Address setup switch	TX lamp turns ON instantly
RS-232C interface	Address setup switch	Displays "[ ]" and RX lamp turns ON instantly
One point I/O	Address setup switch	LED turns ON instantly

Note

(1) Always set up the address one unit at a time.
(2) At the very beginning, send the initialization command "**Z**".
(3) The "**Ac**" command cannot be received during maintenance mode ("**A**" command).

5. Supplement

5.1. ASCII Code Chart

Upper 4 bits

Lower 4 bits	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0				0	@	P	`	p				-	タ	ミ		
1			!	1	A	Q	a	q				ア	チ	ム		
2			”	2	B	R	b	r				イ	ツ	メ		
3			#	3	C	S	c	s				ウ	テ	モ		
4			$	4	D	T	d	t				エ	ト	ヤ		
5			%	5	E	U	e	u			・	オ	ナ	ユ		
6			&	6	F	V	f	v			ヲ	カ	ニ	ヨ		
7			'	7	G	W	g	w			ァ	キ	ヌ	ラ		
8			(	8	H	X	h	x			ィ	ク	ネ	リ		
9			)	9	I	Y	I	y			ゥ	ケ	ノ	ル		
A			*	:	J	Z	j	z			ェ	コ	ハ	レ		
B			+	;	K	[	k	{			ォ	サ	ヒ	ロ		
C			,	<	L	¥	l	|			ャ	シ	フ	ワ		
D			-	=	M	]	m	}			ュ	ス	ヘ	ン		
E			.	>	N	^	n	~			ョ	セ	ホ	゛		
F			/	?	O	_	o				ッ	ソ	マ	゜		

5.2. Send Commands and Respond Commands

	Header	Detail;	Positive response	Negative response
Send	@	Operation specification to the controller	@	n
	Z	Initialization setup	o	n
	Z1	Initialization	o	n
	B1	Automatic completion mode setup	o	n
	M	Operation mode specification	o	n
	Am	Operation mode specification	o	n
	FAm	TW2060 type only, operation mode specification	o	n
	PP1	Block operation instruction with display data specified after pressing the **Fn** key	o	n
	PP5	Rewritable block operation instruction with display data specified after pressing the **Fn** key	o	n
	P1	Block operation instruction	o	n
	P4	Block operation instruction with operation mode	o	n
	P5	Rewritable block operation instruction with operation mode	o	n
	FP1	TW2060 type only, operation instruction	o	n
	FP5	TW2060 type only, rewritable operation instruction	o	n
	H	Completion confirmation	h	n
	D	Operation termination instruction	o	n
	X	Send display code to the batch display	o	n
	Q2		o	n
	T	Signal light ON/OFF	o	n
	G	Barcode reader interface and RS-232C interface	o	n
	Ar	RS-232C interface status check command	a	n
	J	Digital I/O output	o	n
	E	Digital I/O input	e	n
	Ai	Read digital I/O output data	ai	n
	PP2	Display module with mode array	o	n
	P2	Display module	o	n
	FP2	TW2060 type only, light module display	o	n
	A	Maintenance display	o	n
	Az	Module status inquiry	a	n
	Ac	Module's address setup	o	n
Response	t	Barcode reader interface, light module response	O	N (*1)

*1 Only for the board type and RS-232C type. There is no negative response for the Ethernet type.

5.3. PC's Resend Condition

Although the PC has sent a command, there is absolutely no response even after waiting for 5 Seconds for the Ethernet type and 3 seconds for the board and RS-232C types. In this case, the PC resends the command every 5 seconds for the Ethernet type and every 3 seconds for the other types.

Supplement:
It takes time for the controller to process the "**Z**" command (initialization setup) and "**Z1**" command (initialization). For this reason, other than have "receiving timed out," be sure to allow for all commands more than 5 seconds for the Ethernet type and more than 3 seconds for the board and RS-232C types to be processed.

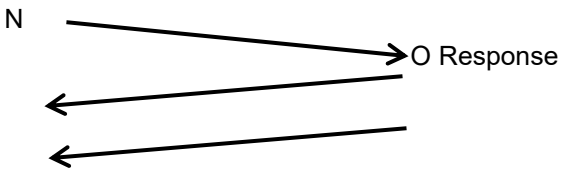
5.4. Condition for the PC's Negative Response

Note: The Ethernet type does not have this function.
If the data sent by the controller cannot be correctly received, send "**N**". The following may be the reason for the negative response.

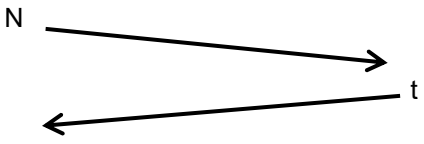
(1) Not enclosed with "**STX**" or "**ETX**" (for the board type).
(2) The checksum is in correct (for the RS-232C type)
(3) There is a mistake in the data length (the data length and the number of data do not match).

Wait at least 3 seconds to determine that the command was not correctly received. If the "**N**" is sent, there are 2 kinds of sequences and 3 reasons.

(1) If this response is a response that has not been processed, the earlier one is considered as a communication problem and processes the current one.
(2) If this response has already been processed, it concludes that somehow it received garbage and resumes.



(3) If it is a "**t**" command, the PC thinks that the previous "**t**" command was a communication error and processes the current "**t**" command.



In another word, if you want to get the controller's response again, send the "**N**" command.

5.5. Relationship between Light Module Part Number and LED Colors

Below is a chart showing various light modules and the colors they can display.

5.1.1 Light Module Color Chart

		red	green	blue	orange	white	purple	light blue	yellow
TW2xxxR		○	–	–	–	–	–	–	–
TW2xxxG		–	○	–	–	–	–	–	–
TW2xxxB		–	–	○	–	–	–	–	–
TW2xxxBi		○	○	–	○	–	–	–	–
TW2xxxD		○	–	–	○	–	–	–	○
TW2xxxF		○	○	○	○	○	○	○	–
TW2xxxRG		○	○	–	–	–	–	–	–
TW5040R		○	–	–	–	–	–	–	–
TW5040G		–	○	–	–	–	–	–	–
TW5040Bi		○	○	–	○	–	–	–	–
TW2xxxSR		○	–	–	–	–	–	–	–
TW2040PRG-22		○	○	–	–	–	–	–	–
TW2928		○	○	○	○	○	○	○	–
SW1xxxR		○	–	–	–	–	–	–	–
AW2xxxF		○	○	○	○	○	○	○	–
DW2xxxF		○	○	○	○	○	○	○	–
NW294x		○	○	○	○	○	○	○	–
NW297x		○	○	○	○	○	○	○	–

Below shows light module part numbers and corresponding LED colors

5.5.2 Light module LED colors specified with the “Am1”, “Am2”, “Am3”, “Mode Array” “m1”, “m2”, “m3”, “PP509” commands

LED			TW2xxxR TW2xxxSR SW1xxxR	TW2xxxG TW5040G	TW2xxxB	TW2xxxBi	TW2xxxD	NW294x NW297x TW2xxxF AW2xxxF DW2xxxF TW2928	TW2xxxRG	TW5040R	TW5040Bi	TW2xxxSG	TW2xxxSY	TW2040PRG -22	
2	1	0													
OFF	OFF	ON	off	off	off	off	off	blue	off	red(lower)		off	off	red	green
OFF	ON	OFF	off	green	blue	green	yellow	green	green(lower)	off	green	off	off	red	green
OFF	ON	ON	off	green	blue	green	yellow	light blue	green(lower)	red(lower)	green(upper) orange(lower)	off	off	red	green
ON	OFF	OFF	red	off	off	red	red	red	red(upper)	red(upper)		green	yellow	red	green
ON	OFF	ON	red	off	off	red	red	purple	red(upper)	red	red	green	yellow	red	green
ON	ON	OFF	red	green	blue	orange	orange	orange	red(upper) green(lower)	red	orange(upper) green(lower)	green	yellow	red	green
ON	ON	ON	red	green	blue	orange	orange	white	red(upper) green(lower)	red	orange	green	yellow	red	green

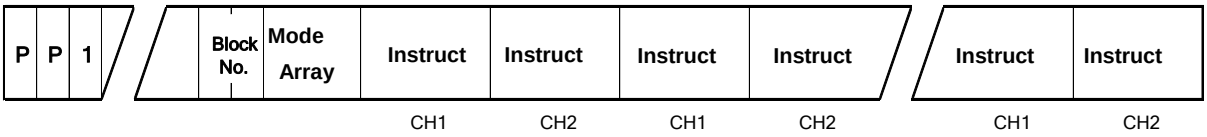
4.5.4 Chart on the light module LED colors with the M, P4, P5, P6 commands

indicated color	TW2xxxR TW2xxxSR SW1xxxR	TW2xxxG TW5040G	TW2xxxB	TW2xxxBi	TW2xxxD	NW294x NW297x TW2xxxF AW2xxxF DW2xxxF TW2928	TW2xxxRG	TW5040R	TW5040Bi	TW2xxxSG	TW2xxxSY	TW2040PRG -22	
00: LED2, 1, 0	red	green	blue	orange	orange	white	red(upper) green(lower)	red	orange	green	yellow	red	green
10: LED1	off	green	blue	green	yellow	green	green(lower)	off	green	off	off	red	green
01: LED2, 0	red	off	off	red	red	purple	red(upper)	red	red	green	yellow	red	grenn
11: LED2, 0	red	off	off	red	red	purple	red(upper)	red	red	green	yellow	red	green

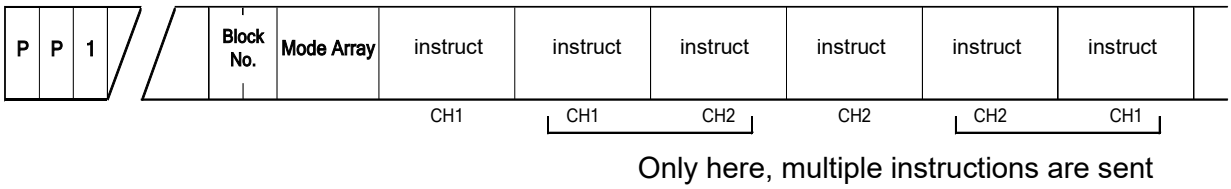
5.6. Optimization of Speed for the Board Type Controller

Note: This function does not apply to the Ethernet type or RS-232C type.

The board type has 2 connecting ports for the modular cable. This means operation instructions can be sent using these 2 channels. However, there are conditions for sending multiple operation instructions. Using the operation instruction commands other than the "**L**" command, arrange the command so that two adjacent light module addresses belong to different channels (channel 1 or channel 2, not to the same channel) and send the command to the controller to obtain optimal speed.

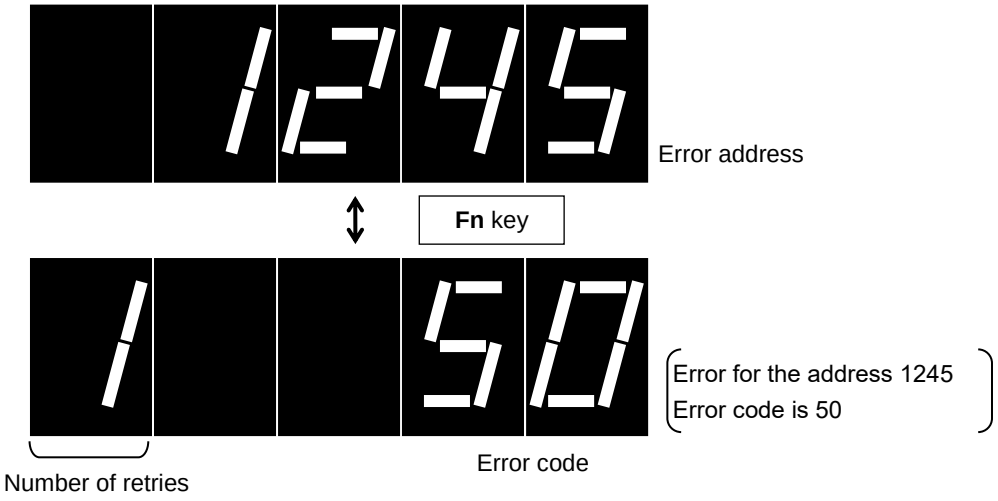


The question is what to do with the Block No. Specify "**00**" to control the entire operation as 1 block, and the completion is confirmed with "**H00**". The PC is aware of the block and zone but not the controller. This means the automatic turning ON/OFF function cannot be used, and up to 5,120 light modules can be treated as 1 Block.



5.7. Number 0 Light Module

Number 0 light module is usually a 5-digit light module with 0 address setting. This type of light module can be connected to any junction box for module error monitoring.
When the operation instruction command or barcode reader interface “Open” command is received from the PC, the controller monitors these addresses. When an error occurs, the error module address and channel error code will be shown on the number 0 light module.



5.8. Error Code Table of Number 0 Light Module

Code	Detail
1	Stopped for an instant
2	BCC error after trying to resend
50	No response (disconnected or damaged)
51	No response (framing error or overrun error)
7	・ More than 2 operation instruction commands are transmitted to the same light module (redundant transmission) ・ Operation instruction command cannot be received.
80	・ Data is received when not expected. ・ No address information
81	・ Data is received when not expected. (framing error or overrun error) ・ No address information

Board type only

5.8.1 Connection Method

Connect Number 0 light module to any junction box. It can also be connected to a channel currently not used. Make sure that the address of Number 0 light module is 0 (zero).

Note

On how to troubleshoot and investigate, see the TW and JW series hardware manuals about troubleshooting and testing.

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[**Inquiries**] Please send inquiries regarding the specifications of L-PICK light module series, the content of the manual, or questions on this manual by FAX or E-mail to AIOI・SYSTEMS CO., LTD. info1@hello-aioi.com

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